

Artificial Intelligence

for the United Nations 2030 Agenda

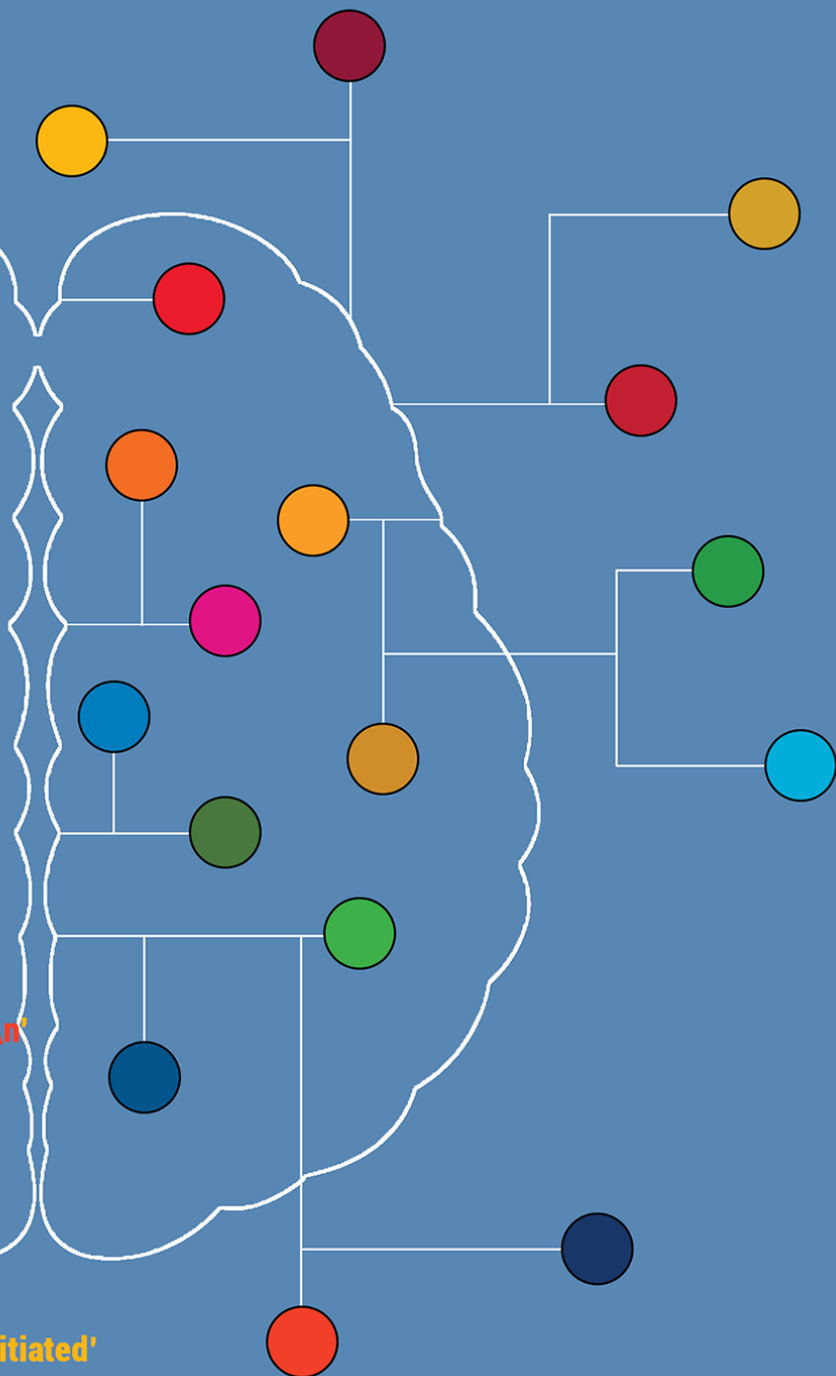
```
def sustainable_development (AI, Innovation,
Responsibility, levelA, levelB, levelC, limit):
    Using as inputs the Artificial Intelligence
    Technology (AI), the existing Innovation and
    people's Responsibility, as well as the current
    levels of Empowerment (levelA), Equality
    (levelB) and Inclusiveness (levelC) to provide an
    output if humanity is ready for sustainable
    development
    #the goals we want to be achieved

    goals = {'Empowerment' : levelA,
             'Equality' : levelB,
             'Inclusiveness' : levelC}

    #check Responsibility first
    if Responsibility == 'high' :
        #then check the existing technology
        if AI == Innovation == 'exists' :
            #check goal status
            for goal, level in goal.items() :
                if level >= limit :
                    print(goal, 'is ready! ' )
                else:
                    return goal, 'is not ready yet!
                    Improve and try again\n'

            else:
                return 'The technology is missing\n'
        else:
            return 'Improve and try another time\n'

    #if everything works, WE ARE READY
    return 'Congratulations!
    Sustainable Development is ready to be initiated'
```



Artificial Intelligence for the UN 2030 Agenda

International Environmental Policy Consultancy – Wageningen University and Research
Final Integrated Report - December 17th, 2018

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Disclaimer notice:

The views expressed in this report do not necessarily reflect those of Wageningen University & Research, nor the experts consulted. Experts speak on their own behalf and their views do not necessarily reflect the views of their institutions.

Preface

Written by fifteen MSc students from Wageningen University and Research during the period of November to December 2018, this report serves as a consultancy project for the United Nations Policy Analysis Branch of the Division for Sustainable Development (UN-DSD). The final aim of the report is to provide inputs to the UN High-Level Political Forum on Sustainable Development to be held in July 2019. Throughout the synthesis report and policy briefs hereby presented, the correlation between a ground-breaking technological innovation, Artificial Intelligence, and the realization of the UN 2030 Agenda will be investigated. The policy briefs explore practical applications of this emerging technical field that fit in the nexus of the six Sustainable Development Goals central to the 2019 High Level Political Forum. More specifically, these SDGs are 4, 8, 10, 13, 16 and 17, which respectively deal with education, economic growth, inequality, climate change, peace justice and strong institutions and global partnerships.



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In addition, we want to thank all consulted experts for their contribution in the creation of this document. Without their knowledge, insight and professional opinions this report would not have been possible. After each respective policy brief, readers will find an overview of the experts consulted throughout the data collection phase.

Executive Summary

Together with the three individual policy briefs, this synthesis report provides policy recommendations for addressing the application of Artificial Intelligence (AI) as a tool for the achievement of the Sustainable Development Goals 4, 8, 10, 13, 16, 17. The underlying theme of “Empowering people and ensuring inclusiveness and equality”, focus of the High Level Political Forum 2019, will be reflected throughout the entire document. The policy briefs address AI Geospatial Mapping Systems, the potential use of AI applications for aid-NGOs and intergovernmental collaboration on Responsible AI.

The brief on *AI Geospatial Mapping Systems* introduces Artificial Intelligence as a cutting-edge technology which could support the mitigation of inequalities in the management of renewable natural resources. Forestry, fisheries and agriculture are the core activities addressed by the policy brief as concrete examples on the role of transparency and open-access in empowering local communities. Furthermore, the application of AI Geospatial Mapping Systems greatly strengthens the capacity of international organizations and national governments in tackling pressing challenges such as illegal fishing and logging, as well as soil degradation.

The policy brief on *Artificial Intelligence for Aid-NGOs* describes the potential of AI applications to improve the operations and response of aid-NGOs within the categories of logistics, scenario planning and situational awareness. Furthermore, the brief highlights the importance of data quality and contextual awareness for the use of AI, in addition to providing recommendations towards improving data methodology and management for aid-NGOs.

The policy brief *Towards Intergovernmental Collaboration on Responsible Artificial Intelligence* focuses on governing the societal impact of AI. It calls for global collaboration by aligning the underlying values of existing AI strategies through multilevel and multi-stakeholder dialogue. The brief presents tools for enhancing intergovernmental collaboration and calls for the establishment of an International Declaration on Ethical Principles for Responsible AI. Finally, it provides practical recommendations for its implementation at the national level.

The policy briefs may be considered self-standing. However, it is recommended to follow the proposed outline of the document in order to gain a better understanding of the nexus between Artificial Intelligence and the achievement of the UN 2030 Agenda for Sustainable Development.

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Introduction

“ (...) AI is advancing dramatically. It is already transforming our world, socially, economically and politically. We face a new frontier, with advances moving at warp speed. Artificial Intelligence can help analyse enormous volumes of data, which in turn can improve predictions, prevent crimes and help governments better serve people. But there are also serious challenges, and ethical issues at stake. There are real concerns about cybersecurity, human rights and privacy, not to mention the obvious and significant impact on the labour markets. The implications for development are enormous. Developing countries can gain from the benefits of AI, but they also face the highest risk of being left behind. (...) ”

António Guterres, Secretary-General of the United Nations

Artificial Intelligence (AI) has the potential to fundamentally alter life and society as we know it. Its development generates both intriguing opportunities and worrisome risks. It is the duty of global leaders and the international community to preemptively enhance positive impacts while mitigating negative outcomes with inclusive and representative measures. Due to the innovative and rapidly changing nature of AI, it is imperative to ensure that its application will not exacerbate existing inequalities and injustices, but rather strengthen mechanisms in order to overcome present gaps and empower vulnerable populations.

The responsible introduction of AI into global society, economy and politics calls for comprehensive and unbiased data, transparent and inclusive institutions, political acknowledgment and empowerment of vulnerable populations, especially in developing countries. As collaborative work becomes essential for reaching the effective management of AI, mechanisms of governance at every level, including private and civil society groups, need to cooperate in order to harmonize efforts towards its responsible integration.

This report aims to present the vital measures needed to effectively and ethically incorporate AI into decision-making and optimize its positive impact in relation to the 2030 Agenda for Sustainable Development. It has been structured to guide the reader through the process of conceptually understanding Artificial Intelligence, in addition to grasping its potential as a supporting tool for decision makers.

Furthermore, the synthesis report is a concise portrait of the technology and processes behind AI. The application of AI towards the fulfilment of the 2030 Agenda is heavily reliant on two considerations. Firstly, the dynamics between AI and decision-making. Secondly, meeting the goal of “Leaving no one behind”. It is paramount to account for varying social, political and economic contexts, especially in developing countries. The individual policy briefs are meant to detail the practical implementation of AI across different sectors and in addition to its respective concerns.

Finally, the report will present a segment on potential future developments of AI in relation to the topics addressed under each policy brief.

Interlinkages

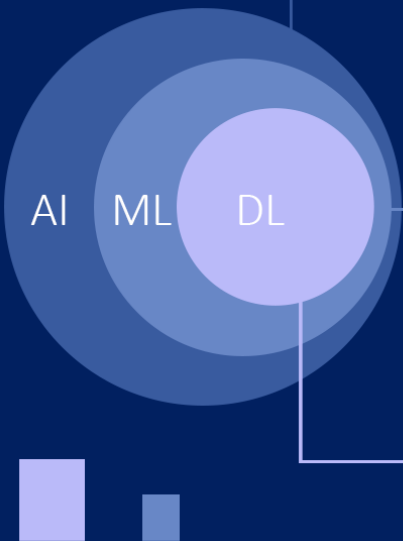
Certain commonalities throughout the three policy briefs enable the establishment of a nexus between them. Firstly, all policy briefs are premised on the recognition of human rights and the need to align AI development with human rights. Additionally, all briefs recognize that certain challenges need to be overcome in order to harness the potential of AI towards the realization of the SDGs. To avoid overly speculating, contemporary challenges are being recognized in the realm of current AI development (in contrast with distant future scenarios) and consist of obstacles regarding inequality, data, access to technology, security and the absence of regulatory frameworks. Moreover, all briefs give recommendations on implementation within a time frame until 2030. Lastly, all briefs are based on both scientific sources as well as expert opinions and are directed towards an audience of senior government officials, policy-makers, as well as any other party interested in the development and governance of Artificial Intelligence.

ARTIFICIAL INTELLIGENCE

Due to its complex and dynamic nature, there is not a universally accepted definition of Artificial Intelligence (AI). Nils J. Nilsson defined it as “the activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment”.

Machine Learning (ML) is the ability of systems to learn by analysing data and identifying patterns, without limited human intervention.

Deep Learning (DL) is a subfield of Machine Learning, loosely modelled after the human nervous systems. The machine learns from vast amounts of data, using sets of algorithms called multilayered neural networks.

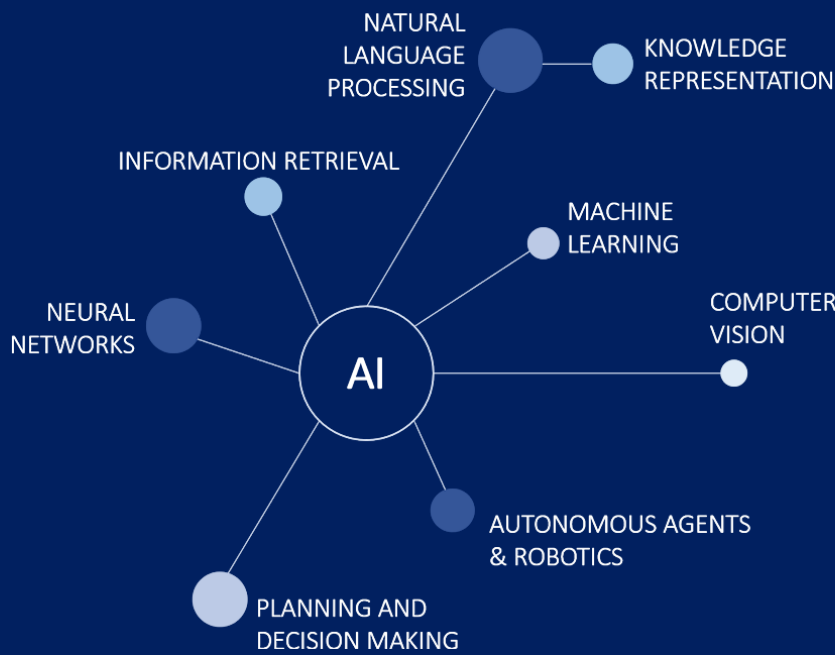


ARTIFICIAL NARROW INTELLIGENCE
TODAY'S AI. UNABLE TO PERFORM OUTSIDE OF THE SINGLE TASK FOR WHICH THEY ARE DESIGNED

ARTIFICIAL GENERAL INTELLIGENCE
NOT DEVELOPED YET. IT REFERS TO MACHINES THAT EXHIBIT HUMAN INTELLIGENCE

ARTIFICIAL SUPER INTELLIGENCE
ANY INTELLECT THAT GREATLY EXCEEDS THE COGNITIVE PERFORMANCE OF HUMANS IN VIRTUALLY ALL DOMAINS OF INTEREST

WE ARE HERE



THE THREE STAGES OF AI

THE SEVEN RESEARCH AREAS OF AI

ACCOUNTABILITY

In an age of increasing free circulation of information, and high-level decisions having national and international impacts, there is an imperative need to strengthen transparency and accountability mechanisms to monitor the responsibility of parties.

ETHICS

The profound complexity of integrating ethical and moral values into AI systems will be one of the greatest challenges of the upcoming years. The use of AI must be assessed in relation to its impact on existing international human rights laws, standards and norms.

PRIVACY

In the context of the so-called information Era, there is the necessity to guarantee citizens the transparent and just management of the data analysed and fed into the algorithms. Legal frameworks around data management are required for the protection of personal information.

BIAS

Bias occurs when the data distribution is not representative enough of the natural phenomenon. Noted should be that Artificial Intelligence algorithms are strictly neutral, and biased outcomes are dependent on biased input. Therefore it is important to prevent biased input from going into an algorithm.

EXPLAINABILITY

Despite the challenges related to the technological progress, one of the most important goals for the advance of AI is to develop systems capable of explaining their rationale for how they perform complex tasks.

SAFETY

Artificial Intelligence development should be based upon common ethical values and should aim at achieving the common good. These systems need to be safe and secure throughout their operational lifetime.

AI for Decision-Making

As the role of AI in society increases and as decision-making becomes increasingly data driven, the plethora of available information poses a challenge to decision makers in terms of data management and analysis. The processing power of AI allows policymakers to analyse data of large volumes and complexity which was previously beyond the analytical capacities of humans. Therefore, information obtained through AI analysis that is responsibly integrated into decision-making can greatly assist in achieving improved equality and quality of life¹.

In the human-AI nexus, people are required to regulate the input of data, guide the training of AI, as well as interpret and validate the outcomes of data analysis in order to conduct more effective and informed decision-making. In this regard, it is absolutely crucial to note that AI is merely a tool that contributes to decisions executed by humans, it is not a solution in itself².

AI applications are dependent on the nature of data submitted. This implies that the input of biased data will result in biased outcomes³. Furthermore, since AI only analyses the data that is provided to it, AI cannot always interpret contextual considerations. It is the responsibility of decision makers using AI to ensure that the methodology behind utilized data is transparent and takes societal, contextual and ethical considerations into account, especially in terms of the privacy of citizens^{4,5}.



The nexus between AI and the SDGs

This section outlines the synergies between AI and SDGs 4, 10, 16 and 17. Although the listed SDGs are by no means the only ones engaged throughout this entire report, these are goals collectively approached by all policy briefs. Additional SDGs addressed only by certain policy briefs will be detailed in the respective sections underneath. Although SDG 13 will not be comprehensively addressed throughout this report, it is an underlying theme in two of the three policy briefs; the same applies for SDG 8 on economic empowerment. *All definitions on SDGs goals, targets and indicators are taken from UN A/RES/71/313^{6,7}.*

SDG 4. Quality education arises as a primary area of interest for AI applications due to its potential to mitigate inequalities within and between countries by enhancing the provision of educational opportunities for all^{8,9}. Nonetheless, relevant parties should be aware of potential negative impacts that AI can have on education, such as standardizing biases on knowledge^{10,11}. All three policy briefs strongly encourage governments and related stakeholders to develop training programs around AI and/or data literacy to provide society the means for responsible AI usage.

SDG 10. The three policy briefs identify core elements concerning inequality within their thematic scopes. There is an overarching synergy between the briefs around equality, empowerment and inclusiveness. To achieve the sub-objectives of each brief, it is essential that all societal actors are enabled to contribute equally in all stages including data input. Consequently, engaging with local actors during the process of analysis is necessary^{12,13}. Furthermore, comprehensive data used for AI coding, governance frameworks established around it and practical applications of AI, such as AI Geospatial Monitoring Systems, strongly depend on the extent of representation of vulnerable groups.

SDG 16. The three policy briefs strongly emphasize the necessity of establishing transparent, effective and accountable institutions to incentivize participatory and responsive decision-making processes¹⁴. Involving non-governmental actors can strongly improve the incorporation of vulnerable groups systematically marginalized by non-inclusive institutions and processes¹⁵. At the same time, direct representation of local communities in decision-making can enhance the distribution of resources essential to their livelihoods¹⁶. Finally, with the introduction of AI to society, institutions around its development need to be strengthened.

Necessary ethical concerns must also be addressed and aligned with existing and new frameworks in order to incorporate efficient accountability measures as well as the representation of vulnerable groups^{17,18}.

SDG 17. Strengthening global partnerships is crucial in the operationalization of the SDGs¹⁹. Therefore, SDG 17 serves as a guideline on secondary methods to implement the recommendations of each policy brief. It is imperative to call for multilateral cooperation that mobilizes financial resources towards establishing the basic infrastructure needed for mitigating the digital divide and the implementation of AI²⁰. Multilateral partnerships should revolve around enhancing South-South, North-South and triangular cooperation, especially in terms of knowledge-sharing and capacity-building for new technologies^{21,22,23}. The three policy briefs converge as promising platforms of collaboration between multiple stakeholders including communities, civil society, governments and private companies.

AI Geospatial Mapping Systems - A Transparent Approach to Natural Resource Management

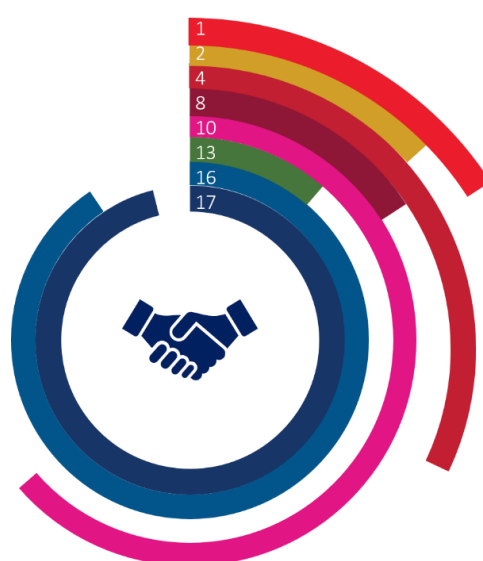
Over the course of the policy brief *AI Geospatial Mapping Systems*, four Sustainable Development Goals are discussed directly, while others are addressed indirectly. The core goals of this document are SDG 16 and 17. Firstly, the importance of accountable and transparent institutions at all levels of governance is key to fully benefit from the integration of AI in the sustainable use and management of renewable natural resources (SDG 16.6)²⁴. Secondly, a multi-stakeholder collaboration is necessary in the creation of publicly accessible data platforms, where input from several actors can contribute to the clear visualization of the state and depletion trend of natural resources (SDG 17). As a result of this cooperation, the decision-making process can be enhanced²⁵, thus enabling policy makers to more effectively implement inclusive policies with the aim to mitigate inequalities in the access to and benefit sharing from natural resources. Equal distribution and implementation of AI has the potential to grant developing countries the possibility of more effectively contributing towards the UN 2030 Agenda, mitigating the existing inequalities between and within countries (SDG 10)²⁶. Another key recommendation in the policy brief is directed towards the equal access to technical education and the promotion of lifelong learning programmes around the use of AI (SDG 4). *Conserving and sustainably using the oceans, seas and marine resources for sustainable development* (SDG 14), is widely reflected throughout the policy brief, especially in relation to targets 14.2, 14.4 and 14.7. Finally, in respect to the sustainable management of forests and land degradation, SDG 15 can be clearly identified throughout the brief.



Artificial Intelligence for Aid-NGOs

The policy brief *Artificial Intelligence for Aid-NGOs* directly targets SDGs 4, 16 and 17. Recommendations related to data methodology and management can increase the availability of high quality, timely and reliable data of developing countries, and improve capacity-building for innovation and technology in developing countries (SDG 17.8, 17.18)²⁷. Recommendations aimed at bundling data methodologies of aid-NGOs facilitate multi-stakeholder partnerships for knowledge, expertise and technology sharing towards supporting the achievement of the SDGs (SDG 17.16). This highlights how open-sourced data methodology and management addresses the need to ensure public access to information and the protection of fundamental freedoms (SDG 16.10)²⁸. By calling for improved data literacy amongst aid-NGOs, this policy brief also contributes towards the number of people skilled in ICT (SDG 4.41)^{29,30}.

Since AI can improve the efficiency and effectiveness of aid delivery, recommendations also indirectly address several SDGs outside the scope of the 2019 HLPF. Through specific AI applications, aid delivery can be made more timely, cheaper and more targeted to the most vulnerable citizens. This can contribute towards reducing the exposure and vulnerability of the poor in relation to environmental disasters and climate-related extreme events (SDG 1.5). Similarly, analytical techniques enhanced by AI could also reduce global hunger by improving access to food by all people (SDG 2.1)^{31,32,33}. Furthermore, it could also improve the effectiveness of programmes aimed at eradicating poverty through more efficient mobilization of resources and improved contextual awareness of impoverished recipients (SDG 1A)³⁴.



Every curve represents the SDG that is related to the number displayed. The length of each curve is dependent on the extent to which the indicators for each SDG are addressed in the policy brief.

Towards Intergovernmental Collaboration on Responsible Artificial Intelligence

The policy brief *Towards Intergovernmental Collaboration on Responsible Artificial Intelligence* targets SDGs 17, 16, 10 and 4. Responsible AI promises to improve inclusive and transparent decision-making in conjunction with the mitigation of AI biases, therewith promoting empowerment and more inclusive decision-making (10.2, 10.3)^{35,36}. In addition, policies on accountability, awareness and capacity-building can lead to the mitigation of economic and social disparities potentially caused by AI developments (10.4)³⁷. By opening an international debate on setting rules surrounding the responsible development of AI, its use and development will be guided by the international community as a whole, instead of by current pioneers of research and development (10.6, 16,7)³⁸. Consequently, SDG 10 can be regarded as the overall aim of the brief. The brief provides a comprehensive list of practical policy recommendations that touches upon most SDGs, particularly SDGs 16, 17 and 4. Enhanced intergovernmental collaboration on the development and use of AI will increase national compliance with international agreements, examples being the Universal Declaration of Human Rights and the SDGs themselves³⁹ (16.10, 16.a and 16.b.). Incentivizing international and national implementation of the core principles of the brief (global collaboration, responsibility and accountability, awareness and capacity-building, data quality and transparency) promotes rule setting on tackling these issues within nations (16.3 and 16.6). Implementing the entailed recommendations will lead to improved international collaboration surrounding the governance of AI, especially in terms of policy and value alignment (17.14). Furthermore, by recommending the establishment of a “UN Responsible AI platform” the brief calls for increased multi-stakeholder collaboration by providing an opportunity to exchange knowledge, best practices and open up discussions on responsible AI (17.16, 17.17). In order to achieve the above mentioned SDGs, improving education (SDG 4) is of key importance as investment in education and capacity-building will lead to the empowerment of vulnerable groups, improved policy-making and decreased inequalities between countries⁴⁰. Although the brief provides recommendations on this topic, it merely views improving SDG 4 as a means to achieve the other SDGs.



Every curve represents the SDG that is related to the number displayed. The length of each curve is dependent on the extent to which the indicators for each SDG are addressed in the policy brief.

Leaving no one behind

“Leaving no one behind” is the overarching goal for the 2030 Agenda. This report addresses the dynamic and complex nature of the goal by incorporating ‘empowerment, equality and inclusiveness’ into its framework. This involves all nations, people and segments of society, including reaching out to those who are furthest behind as well as respecting the ‘dignity of the individual’⁴¹.

In order to address all vulnerable groups such as children, youth, persons with disabilities, elders, indigenous peoples, refugees, internally displaced persons and migrants, there needs to be a significant focus on the disparities within countries. An emphasis on the participation of all segments of society as agents of change is needed, so that they can in turn lead decent, dignified and rewarding lives in a healthy environment. Furthermore, it is essential that all countries equally share developmental benefits for the growth and progress of all, including Least Developed Countries, Landlocked Developing Countries and Small Island Developing States^{42,43}.

However, despite the call for “leaving no one behind”, inequality and poverty are persistent^{44,45}, in addition access to quality education still varies in differing contexts and is becoming more difficult to achieve for different segments of society. While technological advancement can be seen as a tool to tackle global crises, it is also important to note that there are cases where such changes intensify global inequalities. It is also necessary to address the barriers and limitations of technological advancement, more specifically the drawbacks, opportunities and effects framed under the following themes of infrastructure, education and employment.

Infrastructure

The main infrastructural barriers to technological advancement are unequal access to electricity and internet in countries. Such inconsistencies further perpetuate the gap between rural and urban areas within countries, leaving the former at a larger risk of being left behind.

Internet access remains unattainable and unaffordable to a majority of the world population. To elaborate, approximately 60 percent of the global population does not have access to the internet. This translates into roughly 4 billion people who are offline, 2 billion people without a mobile phone and nearly half a billion-people living out of reach of mobile signal⁴⁶. Despite rapid growth in countries such as India and Indonesia, only one-third of the population actively uses internet, and an even lower proportion has access to fixed broadband⁴⁷.

As a consequence, countries with such infrastructural barriers face problems in terms of accessing AI technologies and its potential benefits⁴⁸. In many cases, developing countries lack not only the resources, but the financial structures needed to access the internet and therefore AI.

Education

Access to quality education is another reason for the technological gap between and within countries. One of the main concerns is deficient investment in Research and Development (R&D). Considering that R&D expenditure significantly influences economic trends, there is a requirement for increased attention and investment in R&D for any kind of technological advancement⁴⁹.

Another concern are insufficiencies in the basic knowledge of digital skills. This can only be achieved if countries emphasize their focus on literacy, numeracy, basic academic, digital and entrepreneurial skills.

These gaps in education among and within countries further exacerbate inequalities, emphasizing the need for them to promote education and training systems for skill development. This can be done through an increased focus on collaboration between policy makers, educators and employers, underlining the importance of quality education.

Additionally, local communities need to be included in the process of technological innovation and the adoption of technologies to local contexts for their empowerment. This would leverage benefits towards all segments of society. An example of this is how the government of South Africa launched a project with The World Economic Forum called “South Africa Internet for All”. The goal was to connect as many people as possible to the internet, especially citizens living in rural areas⁵⁰.

Finally, multi-stakeholder cooperation on a global and local level is an essential step towards bridging the digital divide⁵¹. Civil society organizations can play an influential role in facilitating the exchange of knowledge between citizens, the technological community and vulnerable communities in order to empower and engage all segments of society.

Employment

There are a number of large scale impacts of technological advancement on the labour sector. With the advent of AI, the exact effects on the labour market may be hard to predict, however it is possible that jobs and their content will be affected by changes in technology. While the long-term impacts of technological change are unknown, it is assumed that it has the potential to increase productivity and create new jobs, as well as improve standards of living⁵².

However, in the near future, segments of society which have low-skilled jobs are likely to be affected first, further deepening existing economic divides within countries. According to the World Bank, approximately two thirds of all jobs in developing countries are likely to be affected by automation⁵³.

Furthermore, unemployment of disadvantaged communities in the immediate future can consequently push them further behind and contribute to increasing inequality within countries⁵⁴.

Infrastructure, education and employment are some of the core themes in the discussion of bridging the digital divide. Challenges need to be tackled and opportunities need to be highlighted in order to make sure no one is left behind in the emergence of new technologies, specifically Artificial Intelligence.

Policy Briefs



AI Geospatial Mapping Systems

A Transparent Approach to Natural Resource Management



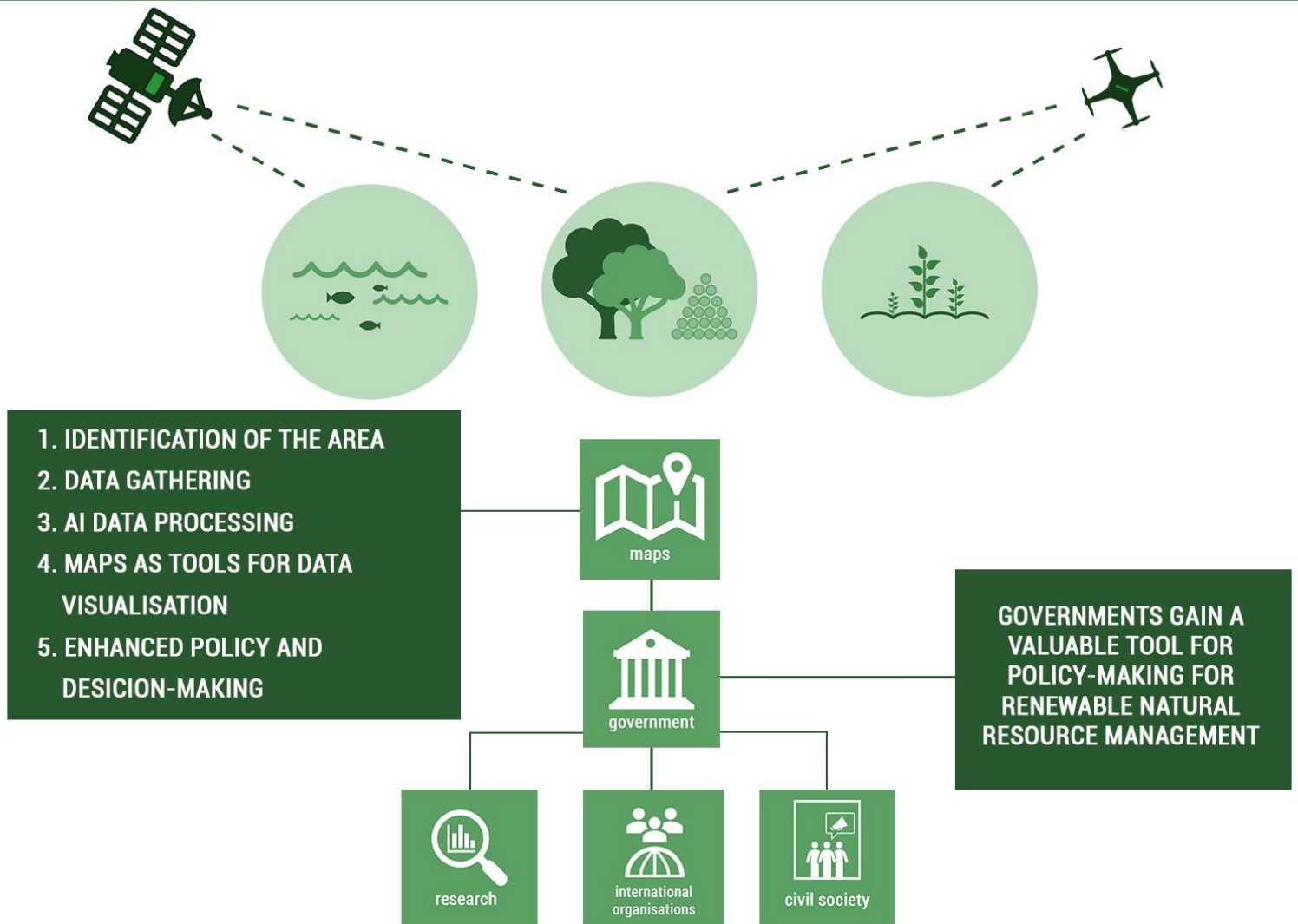
Artificial Intelligence for Aid-NGOs



Towards Intergovernmental Collaboration on Responsible Artificial Intelligence

AI Geospatial Mapping Systems

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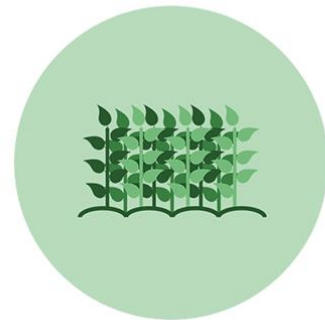
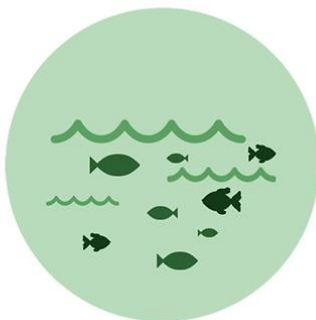


OPEN-ACCESS

TRANSPARENCY

EDUCATION

TRAINING



AI Geospatial Mapping Systems

A Transparent Approach to Natural Resource Management

Key Messages:

1. The unsustainable use of renewable natural resources has become an increasingly pressing issue in Least Developed Countries, Landlocked Developing Countries and Small Island Developing States.
2. Current technological breakthroughs and political drive invigorate the integration of Artificial Intelligence Geospatial Mapping Systems into decision-making on the sustainable use of renewable natural resources by providing scientific and up-to-date information.
3. The rapid development of Artificial Intelligence (AI) facilitates the enhancement of already existing Geospatial Mapping Systems by using Machine Learning algorithms in the image recognition process for faster and more efficient information gathering and analysis.
4. Artificial Intelligence Geospatial Mapping Systems provide a clear picture on the state of resources to users, policy-makers and other interest groups to support addressing inequalities over their access and use, thus contributing to sustainable management thereof. Fisheries, forestry and agriculture are used as illustrative cases in this policy brief.
5. Due to the unprecedented nature of Artificial Intelligence Geospatial Mapping Systems it is crucial to collectively develop transparent and accountable institutions, as well as participatory governance frameworks for the regulation and control over the application and outcomes thereof.

Introduction

Mismanagement of renewable natural resources, particularly in activities such as fisheries, forestry and agriculture, has steadily increased in recent years, contributing to their scarcity^{1,2,3}. As a result, this dynamic has exacerbated competition over the access to and benefit sharing from these activities, particularly in Least Developed Countries (LDC)⁴, Landlocked Developing Countries (LLDC)⁵, and Small Island Developing States (SIDS)⁶. *Mismanagement* comprises misuse and overexploitation of resources, which prevents equal access and benefit sharing as well as the sustainable use.

Fisheries

The constant intensification of human activities and the urgent challenges posed by climate change severely threatens the delicate ecosystem balance of maritime fisheries⁷. The use and sharing of marine biological resources have shaped national and local stability throughout history⁸. Lack of basic capacities and differences in the implementation of technological instruments between and within countries perpetuate already existing inequalities in the access and benefit sharing of activities carried out in the oceans. Lower income nations are unable to efficiently compete in the global fishing market, indicated by the fact that they only account for 3% of trackable industrial fishing⁹. Furthermore, overexploitation and illegal, unregulated, and unreported (IUU) fishing have become pressing matters for governments and international organizations. Various sources estimate that the amount of illegally caught fish ranges between 11 and 26 million tonnes of fish¹⁰.

Forestry

Forests are severely threatened by deforestation due to forestry activities, shifting agriculture and wildfires¹¹. Illegal logging, as a forestry activity, by both companies and local communities currently accounts for at least 50% of all forestry activities in key producing tropical forests (examples being those of the Amazon Basin, Central Africa and Southeast Asia) and at least 15% of all timber traded globally¹². This is due to increased market demand for forest-based products and services in addition to a lack of control over ownership¹³. Illegal logging has significant environmental, social and economic consequences. Forests also provide supporting, provisioning, regulating and cultural ecosystem services which are fundamental¹⁴ to local communities, indigenous peoples and other vulnerable groups. Illegal logging can thus pose a threat to their livelihoods¹⁵, undermine tax revenues for developing countries and force law abiding companies to lower the price of timber products to stay competitive in the market¹⁶.

Agriculture

The intensification of agricultural activities, particularly through the expansion of industrial farming, has accelerated land and soil degradation¹⁷. Researchers estimate that the loss of arable land each year ranges between 5-6 million hectares¹⁸. Furthermore, it has exacerbated water scarcity and groundwater depletion due to increasing water demand¹⁹. Globally, agricultural activities account for more than 70% of total freshwater use²⁰.

In LDCs and LLDCs, agriculture is the most vital economic sector²¹ with the livelihoods of more than 2 billion people reliant on smallholder farms²². As a result, land and soil degradation considerably undermine agricultural productivity and food security²³.

The Innovative Potential of AI

Geospatial Mapping Systems is a type of analysis which gathers, displays and processes spatial imagery derived from aero spatial monitoring devices - such as satellites, small aircrafts, drones - as well as existing maps. Integrating AI in these already existing Geospatial Mapping Systems allows users to analyse vast amounts of data in a faster and more efficient manner²⁴.

AI models are trained by spatial imagery inputs provided by aero spatial monitoring devices. The amount of information provided by these devices is far too large for humans to process and analyse²⁵. Therefore, by implementing Machine Learning algorithms inputs can be rapidly interpreted, which reduces the analysis process and provides more precise and quick information²⁶. Once data is processed through Machine Learning algorithms and validated by specialists, a detailed land cover map is developed, creating an ever-more precise visualization of an examined area²⁷.

While acknowledging the advantages of this innovative technology there are some limitations that need to be accounted for. Since AI algorithms are trained on data submitted by developers, there is potential for data to be biased, which can hinder the entire Machine Learning process. Recognizing the risk of repetition and amplification of human biases²⁸, it is of the utmost importance to consider necessary measures around the regulation and enabling of AI Geospatial Mapping Systems from different perspectives:

- Existing legal frameworks on the use of aero spatial monitoring devices have to be standardized under an international guideline to align efforts towards transparency and open-data.
- To further strengthen transparency in the collection and handling of data, overarching institutions responsible to monitor the process are necessary.
- It is crucial that national and regional mechanisms are established to ensure that all levels of society are incorporated into the debates and decision-making.

The possibility of applying AI to existing mapping systems has already proven to be a promising tool in the management of renewable natural resources. Despite the current difficulties in predicting future developments,

trends show that Machine Learning will soon be able to process complex sets of data and accurately create forecasting models to improve the implementation of spatial planning. If efficient regulatory mechanisms are set in place pre-emptively, this development could further enhance effective policy-making, the promotion of sustainable practices around the use of natural resources and equality for users and beneficiaries thereof.

Transparency at sea

High quality and updated maps designed to track vessels' activity could provide decision makers and local communities a remarkable tool in the use of renewable maritime resources²⁹. Modern developments in AI technologies allow models to monitor vessels through satellite imagery, enabling more effective analysis of sailing and fishing patterns³⁰. Transparency, especially in areas beyond national jurisdiction, is crucial in tackling pressing issues such as IUU catch and overexploitation in open-ocean and deep-sea fisheries, which heavily threaten entire ecosystems³¹. Current Monitor, Control, and Surveillance (MCS) mechanisms can be greatly enhanced by AI through faster and more efficient data processing systems provided by the integration of tracking sensors and aerial imagery³².

At the moment, multilateral collaboration between governments, academia and non-governmental organisations has enabled the creation of open-access and transparent platforms³³, where vessels identification signals (Automatic Identification Systems) are gathered and processed through AI algorithms³⁴. The availability of this information will be beneficial to a wide spectrum of actors. Companies operating legally can present a traceable and sustainable product to their customers. Governments will be able to strengthen their current monitoring mechanisms and prevent the overexploitation of fisheries³⁵. Furthermore, by analysing activities at sea, policy makers and wildlife conservation organisations will be able to more effectively ascertain which maritime regions are severely threatened by anthropogenic activities. The designation of well-governed Marine Protected Areas (MPA) and the promotion of equitable access to the resources therein have proved to be a valuable tool in strengthening local communities' rights^{36,37,38}, allowing them to thrive and compete in an increasingly competitive market.

It is important for governments to recognize the need to standardize regulation around MCS mechanisms, both in the high seas and coastal maritime areas. World nations are preparing an internationally binding treaty under the UN General Assembly regarding biodiversity³⁹ as a reaction to the necessity of strengthening the protection of high seas from overexploitation and IUU catch, as a

further MCS mechanism besides MPAs. On a smaller scale it is also important to regulate and protect small and artisanal fisheries. While MPAs play a big role in allowing specific fish stocks to replenish over time, there is also a need for governments, especially at the local level, to establish tracking mechanisms for small vessels and to promote the use of information derived from AI Geospatial Mapping Systems, in order to build capacity for better and more effective sustainable fishing practices.

Mapping the forest

Together with the previously mentioned application of AI Geospatial Mapping Systems, forest monitoring and governance can drastically improve with the implementation of this innovative technology. Through satellite imagery and Machine Learning algorithms, researchers and policy makers can estimate tree cover loss over time as well as track changes in canopy density⁴⁰. However, not only satellites are involved in AI Geospatial Mapping Systems for forests. In fact, specific areas that require immediate attention are monitored through mapping-drones, which provide high quality resolution maps of targeted zones. As a further development of these systems in the near future, the use of radar mapping tools will allow users to avoid disturbances in pictures created by clouds or other atmospheric obstacles⁴¹.

In order to tackle the social and economic consequences of illegal logging, there is an urgent need for governments in key producing countries to improve the legality and transparency of the timber trade, and to track illegal timber at the market. Governments are making efforts towards the regulation and equal sharing of the benefits of logging^{42,43,44,45}, however further progress can be made. Comparing information regarding logging licences with the data provided by AI Geospatial Mapping Systems will enhance the transparency and real-life monitoring of logging and timber products. Through a transparent and publicly accessible database, local communities can better visualize the impact of forest activities in their surroundings. In addition, companies that are operating legally can present a more sustainable and traceable product, which will give them a market advantage.

Better data, better harvest

AI Geospatial Mapping Systems can be a ground-breaking tool to support farmers through basing agricultural governance policies on interpreted information. Smallholder farmers, who greatly depend on land as an essential livelihood asset are currently being challenged

by low productivity rates due to increasing soil infertility and water scarcity.

Mapping and processing data on the state of soil, water and crops can reduce scarcity while boosting the productivity rate of farmers. Through aerial imagery, vast areas of landscapes can be mapped and monitored, which provides data on soil fertility and on the current state of land degradation⁴⁶. Inspection of access to and availability of water sources can become more efficient and accurate by mapping water infrastructure and usage⁴⁷. Additionally, drone monitoring can support decisions in crop cycle management towards enhancing productivity^{48,49}. Information it provides on different indicators can assist early season operations by determining the optimum amount of seeds for crops, monitoring crop development during the growing phase and efficiently managing harvesting plans⁵⁰.

AI Geospatial Mapping Systems can offer an efficient and affordable approach for LDC, LLDC and SIDS governments to address unsustainable resource usage by facilitating better policy and decision-making in agricultural land and water management. A complete visualization of soil quality on a national scale simplifies land conservation and restoration planning, and enables monitoring the impact of implementation over time. Similarly, governments can easily detect inefficiencies in water use and structure a national water waste reduction strategy. Policies revolving around precautionary water allocation can be meticulously designed ahead of dry seasons.

In order to enhance the productivity of farmers while reducing scarcity in renewable natural resources, collaboration between agricultural and environmental authorities is desirable. The successful implementation of policies addressing sustainable intensification⁵¹ can lead to improved food security.

Farmer cooperatives have proven to empower farmers and strengthen their economic competitiveness. Functioning as sectoral knowledge-sharing platforms and providing members better access to information, cooperatives are a trusted entity and source of reference⁵², that can be an important channel⁵³ to translate the results of regional mapping and monitoring to farmers in terms of concrete suggestions and best practices. In this sense, the use of AI Geospatial Mapping Systems can be applied from high level governance to day-to-day agricultural management.

Challenges

Inasmuch the use of AI Geospatial Mapping Systems relies upon an innovative technology, potential challenges need to be promptly addressed before they outweigh advantages and jeopardize objectives. If the access to and use of this technology are limited to a few powerful actors, there is a risk that benefits thereof will be unequally divided, which might exacerbate inequalities within and across societies⁵⁴. Therefore, it is advisable for governments to take part in collective action in order to support existing open-access databases for the optimal deployment of AI Geospatial Mapping Systems across countries.

Moreover, AI integrated Geospatial Mapping Systems are the perfect platform to advocate for unifying international and national governance efforts on the use and monitoring of data for the purpose of improving mapping and policy making. This technology relies heavily on the availability of data to generate maps and spatially visualise information. Data sharing initiatives from governments and international organizations providing free satellite imagery already exist⁵⁵, making it possible for all governments to access this information. Nonetheless, to ensure that all segments of society are reached, the spread of information needs to be institutionalised across authorities and local governments, in addition to being fostered through multilateral collaborations.

Enhancing factors

Technology

Acquiring and using drones as well as AI technologies is becoming more affordable and cost-effective than ever^{56,57}. However, inequalities in the application and distribution of this technology are not only limited to the monetary value of devices, but also to the required human resource capacity. Sustaining the cost of human expertise can be a constraint in the development and use of this technology.

Education

To ensure sufficient carrying capacity for implementation, education is one of the key measures in tackling the unsustainable use of renewable natural resources and incorporating vulnerable populations into the sharing of benefits^{58,59,60}. Education is a promising means by which to tackle social inequalities⁶¹. To avoid the risk of an uneven application of AI Geospatial

Mapping Systems widening the gap between and within countries, governments need to make preliminary investments in educational programmes on AI, in addition to fostering international collaborations and exchange programmes on the development of this technology⁶². Furthermore, training needs to be provided in order to enable local governance actors to use technology interfaces and to understand information coming from AI analyses, so that they can facilitate the process of designing policies around the sustainable and inclusive use of natural resources.

Transparency

In order to create an international open access database to share information on the use of renewable natural resources, political commitment is needed from all parties involved. To optimally establish of this tool in the application of science-based data for decision-making, there is an imperative need to bring together different users and stakeholders under an international umbrella that ensures the transparency and quality of processed data. An international framework on the use of AI Geospatial Mapping Systems can help guarantee transparency in data processing, while building trust and ensuring equal terms between participating countries at the same time.

RECOMMENDATIONS

-  Improve coordination between international organizations, governments, and civil society in the creation of an open-access database where continuously updated maps are essential tools in the monitoring of specific areas.
-  Establish governance frameworks and institutions to ensure transparency of data gathered through AI Geospatial Mapping Systems.
-  Promote educational programmes on the use and programming of Machine Learning algorithms in order to engage youth in the development of AI technologies.
-  Encourage the use of AI Geospatial Mapping Systems by local governance actors in decision and policy making around renewable natural resources through specific professional training.

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Annex I: Consulted experts

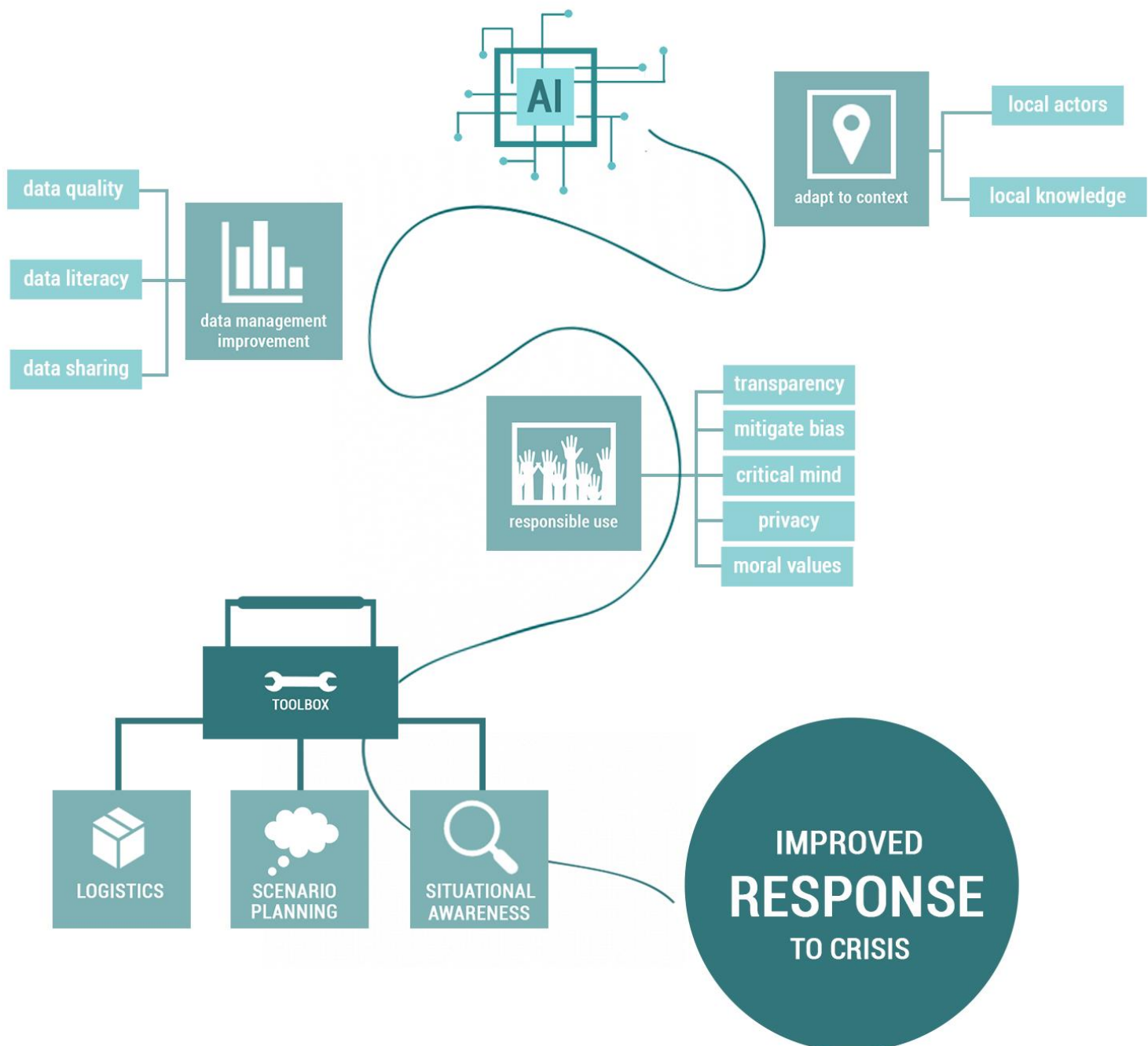
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Name	Institution	Position	Interview Date
Sarah Bladen	Global Fishing Watch	Communications and Outreach Director	4-12-2018
Prof. Dr. Tibor Bosse	Radboud University Nijmegen	Professor Communication Science and Artificial Intelligence	28-11-2018
Dr. Ranveer Chandra	Microsoft	Principal Researcher	10-12-2018
Maartje Holtslag	ESRI	Young Professional Software Development	6-12-2018
Dr. Eng. Aad Kessler	Wageningen University and Research	University Lecturer	29-11-2018
Anas Khasawneh	UNDP Jordan, GEF Small Grants Programme	National Coordinator	28-11-2018
David Kroodsma	Global Fishing Watch	Director of Research	12-12-2018
Dr. Kyungsun Lee	X-Grant Project	Postdoctoral Research Associate	7-11-2018
Dr. Dik Roth	Wageningen University and Research	University Lecturer	26-11-2018
Camiel Verschoor	Birds.ai	Founder and CEO	29-11-2018
Dr. Hans-Peter Verschoor	Wageningen University and Research	University Lecturer (head)	22-11-2018
Mikaela Weisse	Global Forest Watch	Manager	7-12-2018

Artificial Intelligence for Aid-NGOs



improve operations



Artificial Intelligence for Aid-NGOs

Key Messages

1. There are a variety of Artificial Intelligence (AI) applications that show potential to be implemented to improve logistics, scenario planning and situational awareness of aid-NGOs.
2. Since AI applications require quality data and contextual awareness, feasible implementation within the aid community requires improvement in data quality, data collection and management.
3. Given the nature of AI, the limited experience of society with the technology, and the complexity of issues that aid-NGOs are dealing with, applications of AI cannot be implemented without addressing biases in data collection and ensuring privacy, trust, and transparency regarding the use of AI.

Introduction

The frequency and severity of worldwide crises is increasing, from natural disasters such as floods to global issues such as food security¹. These developments underline the increasingly important role that humanitarian as well as development NGOs (could) play, and why it is crucial they make timely and accurate decisions. There are significant differences regarding the aid delivered by development and humanitarian NGOs (both encompassed under the term aid-NGOs), yet they are looking for ways to deliver aid in a more effective and efficient manner. In order to achieve this, aid-NGOs must continuously improve their decision-making process and operational strategies, especially while facing exacerbated threats due to climate change.

Technological innovations such as AI applications can assist aid-NGOs in their decision-making processes². More specifically, AI can be used by aid-NGOs in a variety of ways, such as finding the fastest route to deliver aid or identifying the most vulnerable people during disasters. With this, AI also facilitates the implementation of the Sustainable Development Goals (SDGs)³.

Therefore, this policy brief is written for policy makers, researchers, and aid-NGOs alike to provide orientation on how AI applications could be applied for aid-NGOs.

The potential implementation of AI applications within the aid community requires the careful management of technical and societal considerations that are associated with potential benefits and drawbacks of utilizing AI technology. More specifically, poor data quality will cause the potential benefits of AI to be delayed, misused, or not implemented at all⁴. Feasible and responsible use of AI applications requires significant improvements in aid-NGOs' data collection and management.

AI requires data, but good data

AI applications require high quality data, in order to produce inclusive and trustworthy results. A variety of existing data sharing initiatives such as the International Aid Transparency Initiative ([IATI](#)) and the Humanitarian

Data Exchange ([HDX](#)) are already used by a number of organizations^{5,6}.

While such initiatives have contributed to data sharing of aid-NGOs, experts indicate the collected data is often not sufficiently informative, in that they do not measure actual impact. Moreover, datasets are occasionally missing units (individual statistics), are not based on representative samples and are not up to date. Additionally, datasets are poorly structured at times. Furthermore, aid-NGOs often lack the technical expertise or financial capacity needed to address these issues.

AI applications for Aid-NGOs

Aid-NGOs today could choose from several data-oriented software for data collection, visualisation, management and analysis, as well as AI applications. Aid-NGOs often lack the capacity to make informed choices regarding developing technologies, thus adequate expertise and guidance is needed.

Despite the current hype surrounding the capabilities and opportunities that AI applications offer, in many situations, a variety of alternative analytical techniques and human knowledge might offer more viable solutions. Additionally, aid-NGOs frequently operate in regions lacking internet, mobile service or electricity, which can render the implementation of some AI applications unfeasible. Therefore, it is important to further investigate the potential of utilizing AI applications within the aid community.

The potential of AI applications for aid-NGOs

With improved data management, a variety of AI applications have the potential to be used in the near future and can assist with the implementation of the SDGs by 2030. This policy brief serves as an overview of this potential. As reflected in academic literature, professional briefs, and our expert interviews, potential applications of AI technologies can be divided into three categories: logistics, scenario planning and situational awareness.

Category A: Logistics

Logistics, specifically supply chain management, is concerned with the delivery of goods from a point of origin to a destination. Research has estimated that up to 80% of aid-NGO expenditure falls under the domain of logistics^{7,8,9}. Furthermore, improved data management and analysis will have significant potential to increase operational efficiency through using AI. Aid-NGO practitioners in the field confirm that AI applications are mostly absent, and that aid supply chains are currently volatile, unpredictable and slow¹⁰. Private sector applications of AI in logistics have proven that there is potential in improving supply chain management for aid-NGOs¹¹. Considering accessibility and affordability, the following possible applications should be prioritised.

Intelligent Route Optimization

Intelligent route optimization allows real-time routing algorithms to assist in determining the most efficient and cost-effective aid delivery route, thus freeing up more funds. These algorithms could be used in conjunction with satellite maps, social media data and traffic flows, for routes by land, sea and air.

Predictive Risk Management

Predictive risk management can identify (indicators of) risks to aid-NGOs' supply chains in advance, which could help these chains enhance continuity of their supply chain and avoid disruptions in their aid delivery. Through combined Machine Learning (ML) and Natural Language Processing (NLP), it identifies sources of risk by mining real-time data from Internet websites.

Aid-NGO supply chains differ greatly in their properties. For humanitarian aid, the rapid delivery of aid is crucial. However, because humanitarian supply chains have to be set under time pressure, they are particularly prone to inefficiencies, bottlenecks, and mistakes. Additionally, these supply chains are not intended to be implemented in the long term, yet require enormous amounts of aid to be moved at once. This makes dynamic intelligent route optimization using AI have significant potential in the context of a crisis prone future. Developmental supply chains could, similarly to humanitarian supply chains, benefit from intelligent route optimization since they are implemented long-term. Large organizations supplying large quantities of aid, such as food or medicine, would see their transportation costs fall. Such applications could be used in conjunction with predictive risk management in order to ensure supply chain continuity in volatile situations.

Category B: Scenario Planning

Scenario planning is used to analyse the different ways in which situations might evolve, in order to prepare and plan ahead for alternative future developments¹².

Scenario planning is already being used among aid-NGOs, but a variety of aspects can be improved¹³. Recent developments in AI have suggested its potential in enhancing scenario planning by providing both a dynamic analysis of data patterns as well as improving forecasting capabilities.

Scenario Generation & Selection

AI can assist the generation of multiple scenarios for aid delivery by mining essential information from websites in real time using NLP applications. Scenarios that have the highest possibility to describe a future state can then be prioritised, saving time and resources.

Early Warning Systems & AI

The goal of early warning systems in scenario planning is to estimate whether a stable situation is crisis prone¹⁴. Predictive Analytics using ML can enhance such an early warning system by learning which indicators are the strongest predictors of a crisis, in addition to assessing whether current conditions are similar to those of past crises¹⁵. Knowing when a crisis might occur before it does can greatly increase the effectiveness of aid delivery.

Humanitarian aid-NGOs deal with rapidly unfolding crises. Therefore, they need to make certain that any strategy is solidly built and sufficiently flexible to withstand changes in the operational environment. The development actors are required to maintain certain levels of aid in specific sectors, often working under unstable and sensitive conditions. An example of an AI application for scenario planning currently being developed is the Famine Action Mechanism¹⁶. Involved actors utilize AI and ML to build an Early-Warning System which will forecast and estimate food security crises in real-time as well as analyse early signs of food shortages, using indicators for conflicts, natural disasters and crop failures.

Despite opportunities for improvement in scenario planning utilizing AI, it is important to realize that no scenario is an exact prediction of the future. AI applications for scenario planning are still in early stages of development.

Category C: Situational Awareness

Situational awareness is used in order to understand all aspects of a situation or environment, and is especially helpful in making complex and informed decisions in a time sensitive manner¹⁷. Applying situational awareness helps improve contextual understanding and finding the most vulnerable populations. Accurate, instant and complete information that simultaneously bridges both the gap between perceptions of reality on the ground and high up in organizations is essential in improving the efficiency and quality of aid delivery by NGOs. Additionally, such information is crucial for the organization of different sources of aid simultaneously. AI

can play a significant role in improving situational awareness, as it is capable of analysing large quantities of information much faster. There are several possible applications using AI for improving contextual understanding and finding the most vulnerable populations that show potential.

Aid-NGOs have experimented with applications improving situational awareness using ML. Unfortunately, applications for aid-NGOs are currently neither widespread nor feasible for organizations with limited resources.

ML for Priority Assistance

To locate the most vulnerable populations, ML can be used to identify specific cases that require immediate attention. Specifically, ML could process information (e.g. language on social media), to assist in gaining an overview of the affected areas or individuals concerned. It is different from scenario planning, in that it is used to instantly identify the most vulnerable populations amongst a very large and complex set, without necessarily assessing likely future scenarios based on more general data.

Spatial awareness

Situational awareness can be greatly improved with the help of AI combined with satellite imagery. Since satellite imagery is more readily available, it can be used by aid-NGOs to instantly identify affected areas in order to identify which region needs the most support. A drawback of this application is that vast collections of labelled images are required to 'teach' the algorithm how to recognize the patterns of interest. This problem is recently being tackled, however, by using crowdsourced data: an aid-NGO can rely on the support of many people via the internet to label images.

In case of humanitarian disasters, aid-NGOs could make use of social media data to map affected populations, and potentially identify the most vulnerable cases among them. The Artificial Intelligence for Disaster Response ([AIDR](#)) is an existing platform that uses social media posts in crisis situations to create up-to-date maps of affected regions, supporting prioritization of aid delivery to severely impacted areas.

Satellite imagery combined with ML could be used to instantly identify areas affected by a natural disaster or identify human rights violations (e.g. based on livelihood destruction). [Crowdsourced image labelling](#) is used by several aid-NGOs to manually identify patterns of affected areas. Such initiatives could feasibly consider trials to automate the labelling process with the help of ML. This could already considerably accelerate the process of identifying affected areas.

Recommendations

Improve Data Quality

Well-structured data sets without input errors, duplicated or missing data, are a requirement for the successful implementation of AI applications. Improvements in the quality of data should be made prior to AI implementation. Current data is not sufficiently informative. Thus, aid-NGOs should reinforce collaborative efforts to improve their methodologies in order to collect more inclusive and coherent data of better quality.

Coordinating methodologies on a global scale for all aid-NGOs simultaneously is, however, not feasible. There are simply too many variables that would need to be measured. NGOs should form groups in which methodologies are coordinated. Humanitarian aid-NGOs could for instance coordinate methodologies through the UNDAC (United Nations Disaster Assessment and Coordination), which already has a data collection structure for disaster relief. If an UNDAC structure is not available, e.g. in development aid, bundling methodologies based on the humanitarian [UN Cluster Approach](#) and geographical regions could assist aid-NGOs in enhancing their data collection and management. Practically, aid-NGOs should make lists of variables measured by every aid-NGO in their subgroup to make impact evaluations as informative as possible, while simultaneously enabling the triangulation of results.

During impact evaluations, aid-NGOs must ensure that variables added are as inclusive as possible, taking cultural and societal considerations into account. Additionally, data of higher quality can enhance the triangulation of observations among different aid-NGOs, thus allowing more effective responses. To improve effectiveness and inclusiveness, aid-NGOs should involve local actors such as universities or local organizations in the data collection and analysis process. Such actors could accelerate relevant variable selection and introduce new variables to look at. The United Nations could further act as a facilitator for regional and cluster-based coordination, by bringing relevant aid-NGOs together.

Mitigating Biases

Data and data sets are not objective¹⁸. People set the variables, collect the data, define its meaning through interpretations and draw inferences from it. Taking hidden biases in both the collection and analysis stages into account, is as important to consider for data quality as the selection of variables itself. Even high-quality data is not necessarily an accurate representation of society. There are cases where no, too little or too much input is coming from particular demographics or communities.

For aid-NGOs, it is also important to note that people familiar with the local context often have valuable, qualitative knowledge that might not be quantifiable. Therefore, an algorithm may be unable to integrate particular information, leading to biased outputs.

Recognizing the biased nature of data is a step towards adopting a healthy and critical perspective concerning the outputs of the AI applications. Most importantly, representative datasets should be used in order to capture the relationships that may exist within the datasets of the input, as well as between datasets and output results.

Responsible use of AI applications

Privacy

Data management preceding the use of AI applications, requires data privacy¹⁹. Sensitive data inputs should not be traceable to individuals or demographics. Where individual variables are neutral, it is important to note that a set of variables can enable somebody to deduce that a group of data records is associated with a certain region or group. Furthermore, data sets used to train ML applications should be in line with data privacy regulations (e.g. General Data Protection Regulation (GDPR) by the European Union).

Trust

The outcome of an algorithm is often as biased as the input of data itself. Therefore, applying the outcome to support a decision should be met with healthy, critical reflection. The use of AI needs a fair amount of mentoring. It is important to avoid relying solely on an AI application for making decisions²⁰. The outcome must be continually evaluated under scientific knowledge, local knowledge, past experience and common sense. This incorporates accountability and the need for representation of moral values that are present in the context of a situation²¹.

Transparency

Responsible AI use requires data transparency. This is a core principle of data protection since people have the right to know if and to what extent their personal data is collected, used or processed. It is also essential that aid-NGOs deliver feedback about the use of collected data and the results of any conducted research to the local people. This process can empower and benefit local communities as well as foster a collaborative relationship.

Looking forward

It is essential for the workforce of aid-NGOs to improve their data management skills to successfully work with AI applications. Data literacy can be achieved through training programs for the workforce, potentially funded

by donors and in collaboration with existing infrastructure, an example being the Active Learning Network for Accountability and Performance in Humanitarian Action (ALNAP)^{22,23}. Increased data literacy among aid-NGOs will help bridge gaps between aid-NGOs and the technological community. Furthermore, researchers and computer scientists could assist aid-NGOs in data collection and analysis in order to better facilitate AI integration. In the long run, this will enhance data sharing initiatives.

In order to assess the applicability of AI tools in a specific situation, the following two considerations should be taken into account. After a clear problem is defined, whether there is a more efficient and cost-effective alternative to the use of AI should be explored. Furthermore, local contextualities should not be underestimated; AI applications should be tailored to the situation. Individual technologies described above could for example be used to complement each other.

The Hakeem chatbot, for example, is an application developed to assist relief workers in educating young refugees in the local language²⁴. It combines ML and NLP to provide educational advice for online studies, according to users' interests. At the same time, the Hakeem chatbot improves as a 'study advisor', relying on users' feedback. The pilot use of the application with conflict-affected youth in Lebanon, has brought education to individuals that did not have access to these advances before using AI.

Such applications can assist aid-NGOs to be more prepared for future crises and to provide aid more efficiently. This will facilitate aid-NGOs' increasing role in implementing the SDGs.

KEEP IN MIND

-  Where possible, all stakeholders should commit to improving data quality and data sharing.
-  Actors involved should realize implementing AI in the decision-making process must never result in undignified treatment of human lives. Recalling that these decisions concern human lives is essential.
-  Involvement of local actors in the decision-making process, as well as in the implementation of AI applications is necessary to enhance the quality of action and community empowerment.
-  Not every problem is solved with (just) a technological solution. AI is not panacea, but rather one of the tools in the aid-NGOs' toolbox.

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Annex I: Glossary

- **Accountability in Artificial Intelligence:** Who, or what, is held accountable when AI systems make decisions that affect individuals or the society.
- **Artificial Intelligence (AI):** 1. The field of computer science dealing with the ability of a computer program or a machine to think and learn. 2. The ability of a computer program or a machine to perform tasks that mimic human cognition.
- **Aid-NGOs:** A term used to encompass humanitarian and development (aid) Non-Governmental Organizations.
- **Algorithm:** A process or set of instructions and/or rules, that can be used in problem-solving operations and calculations.
- **Bias:** Normatives, stereotypes and opinions from the real-world are encoded and represented within AI systems.
- **Chatbot:** An AI computer system that convincingly simulates human behavior in a conversation via auditory or textual methods.
- **Crowdsourced data:** Information/ input on a particular theme, acquired via the participation of a large group of individuals
- **Data literacy:** The ability to obtain, understand and communicate meaningful information from data, as well as to handle the competencies of working with data.
- **Development Aid:** Delivering long-term aid to address structural problems in developing countries.
- **Humanitarian Aid:** Delivering short-term aid immediately after a disaster to save lives under the humanitarian principles of humanity, neutrality, impartiality and independence.
- **Information (Data) Mining:** The process of generating new information and discovering patterns from large data sets using statistics, Machine Learning and other methods.
- **Intelligent route optimization:** Improving the distribution network to enhance the real time delivery efficiency.
- **Real time routing algorithms:** Algorithms with the ability to handle *real time* network updates that can be used to compute shortest path.
- **Machine Learning:** A branch of the broader AI Technology, that explores the idea of machines processing data and learning on their own to improve their performance on a specific task, without requiring human supervision.
- **Natural Language Processing:** A sub-branch of computer science, information engineering, and artificial intelligence studying the ways computers can process and analyze large amounts of text and audio natural language data (human language).
- **Panacea:** A solution which solves every problem.
- **Predictive Analytics:** A method of examining historical and current facts to formulate predictions about future or otherwise unknown events.
- **Real-Time Data:** Information that is being delivered immediately after collection/generation. The data is analyzed using real-time computing or stored for later or offline data analysis.
- **Triangulation:** The method of collecting, comparing and combining data from two or more sources, in order to validate this information.
- **UN Cluster Approach:** A structuring of the humanitarian sector into eleven clusters such as logistics, food security and education in order to increase coordination between (UN and non-UN) organizations to enhance the humanitarian response to a disaster.

Acronyms

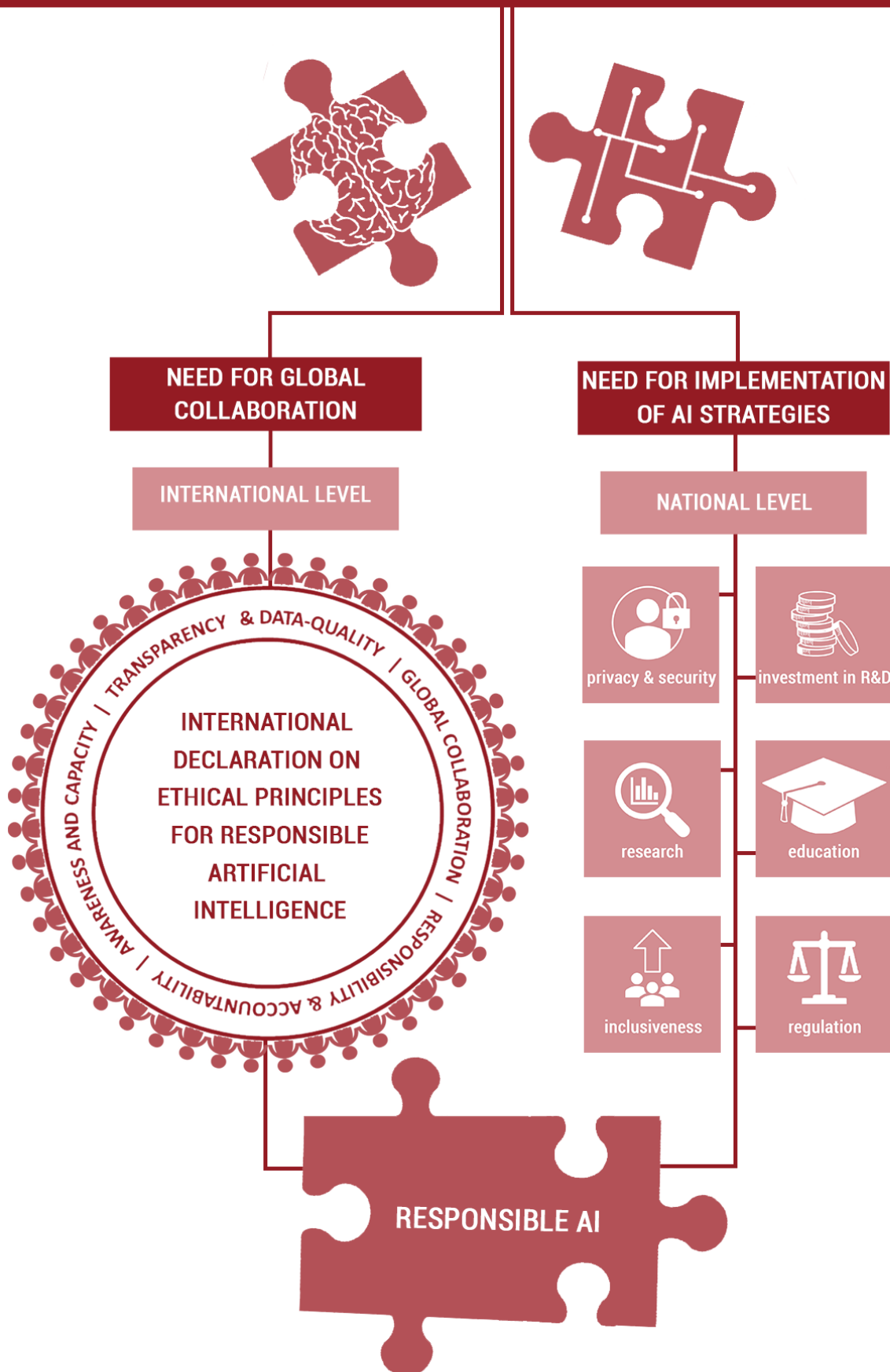
- | | |
|--|--|
| • AI = Artificial Intelligence | • ML = Machine Learning |
| • AIDR = Artificial Intelligence for Disaster Response | • NGO = Non-Governmental Organisation |
| • ALNAP= Active Learning Network for Accountability and Performance in Humanitarian Action | • NLP = Natural Language Processing |
| • GDPR= General Data Protection Regulation | • SDGs = Sustainable Development Goals |
| • HDX= Humanitarian Data Exchange | • UNDAC = Netherlands United Nations Disaster, Assessment and Coordination |
| • IATI= International Aid Transparency Initiatives | |

Annex II: Consulted Experts

We would like to express our gratitude to the people that agreed to be interviewed for this policy brief. Their guidance and feedback have been of crucial importance for the accomplishment of this brief. Furthermore, we are thankful to *dr. Aalt-Jan van Dijk*, *prof. dr. ir. Hein A. Fleuren*, *Dilek Genc (PhD Candidate, International Development)*, *dr. Bram J. Jansen*, *drs. ing. Kenny Meesters*, *prof. dr. ir. Dick de Ridder*, and *dr. Jeroen F. Warner* for reviewing the information included in this brief, according to the field of their expertise.

Name	Institution	Position	Interview Date
Renée van Abswoude	Kuno - Platform for Humanitarian Knowledge Exchange	Intern	22-11-2018
Dr. Aalt-Jan van Dijk	Wageningen University & Research	University Lecturer, Mathematical and Statistical Methods - Biometris	06-11-2018
Dilek Genc, PhD Candidate	University of Edinburgh	PhD Candidate, International Development	15-11-2018
Sam de Greeve	Plan International (Netherlands)	Humanitarian Fundraiser and Programme Officer	26-11-2018
Ankit Gupta	Dasra (NGO, India)	Associate, Systems & Operations	17-11-2018
Dr. Eng. Gemma van der Haar	Wageningen University & Research	University Lecturer, Sociology of Development and Change	09-11-2018
Dr. Bram J. Jansen	Wageningen University & Research	University Lecturer, Sociology of Development and Change	13-11-2018
Daniel Kersbergen	The Netherlands Red Cross, 510 team	Surge Information Management Officer	21-11-2018
Dr. Sanneke Kloppenburg	Wageningen University & Research	University Lecturer, Environmental Policy	14-11-2018
Eng. Kenny Meesters, PhD Candidate	Delft University of Technology	PhD Candidate, Policy Analysis group	28-11-2018
Prof. Dr. Eng. Dick de Ridder	Wageningen University & Research	Professor, Bioinformatics	09-11-2018
Andrea de Ruiter	Ministry of Internal Affairs (Netherlands)	Senior Policy Officer	28-11-2018
Sjaak Seen	Netherlands United Nations Disaster, Assessment and Coordination (UNDAC)	Multidisciplinary Operational Team Leader	14-12-2018
Prof. Dr. Devis Tuia	Wageningen University & Research	Professor, Laboratory of Geo-information Science and Remote Sensing	09-11-2018
Dr. Harwin de Vries	INSEAD Social Innovation Centre	Post-Doctoral Research Fellow	22-11-2018
Prof.Dr. Bartel A. Van de Walle	Delft University of Technology	Professor of Policy Analysis and Head of the department Multi-Actor Systems.	23-11-2018
Dr. Jeroen F. Warner	Wageningen University & Research	University Lecturer, Sociology of Development and Change	12-11-2018
Sven Warris, PhD Candidate	Wageningen University & Research	DLO Researcher, Bioscience	09-11-2018
Titia Wenneker	Giro 555	Online Coordinator	22-11-2018

Towards Intergovernmental Collaboration on Responsible Artificial Intelligence



Towards Intergovernmental Collaboration on Responsible Artificial Intelligence

Key Messages

1. Challenges and opportunities brought about by the emergence of Artificial Intelligence (AI) are of a transboundary nature and call for a global approach. The development of AI needs to be guided in such a way that its benefits are distributed equally within and among countries.
2. The UN should consider aligning existing ethical values of national AI strategies through multilevel and multi-stakeholder dialogue. This will provide input for an *International Declaration on Ethical Principles for Responsible Artificial Intelligence* (IDEPRAI).
3. Responsible AI can be put forward at an international and national level. The UN facilitates a platform for addressing challenges, sharing knowledge and exchanging best practices of AI. An AI task force will be responsible for the national implementation of Responsible AI policy.

This policy brief appeals for global collaboration to work towards Responsible AI. Structured by four core principles, the brief provides practical implementation tools for the UN and national governments to effectively work towards this aim.

Artificial Intelligence (AI) is a cutting-edge technology that promises to improve the wellbeing of humanity¹. However, the concentration of knowledge capital, economic capital and expertise on the development of AI could exacerbate vulnerabilities in economic and social structures^{2,3,4}. Therefore, concerns are being raised as to whether the benefits of AI will be distributed equally within and between countries^{1,5}. Besides that, current global development of AI technology is mainly limited to companies and therewith prevalently reflecting private interests¹. Consequently, the international community faces the challenge of ensuring that technological innovations are in line with the Sustainable Development Goals (SDGs)¹.

In the current governmental playing field, a trend in creating national AI strategies can be observed³⁴⁻⁴⁶. This reflects the acknowledgement of potential dangers of AI developments for society, the need to take governmental responsibility in mitigating these risks and the opportunity to harness the potential of AI⁶. Since most of the expected risks of AI are global, they should be addressed at an appropriate scale⁷. Currently however, global cooperation on Responsible AI is mostly initiated by the private sector^{4,8}. This brief identifies a gap in intergovernmental collaboration⁷. Many countries do not have an AI strategy and countries with AI strategies often lack practical approaches for implementation⁹.

Need for a common approach: A Declaration on Responsible AI

A common vision is important for working towards responsible use and development of AI^{10,11}. This means that humans need to deal with AI in a democratic and socially accountable manner, while guaranteeing human autonomy and values in conformity with international standards (e.g. Human Rights and the SDGs)^{1,10,14}. In terms of governance, this requires connecting common

values from existing strategies and aligning them with Human Rights and the SDGs¹. This brief regards the UN, with its experience in addressing transnational issues such as Human Rights^{12,14} and Climate Change¹³, as an institution capable of playing a leading role in intergovernmental collaboration on Responsible AI.

To organise intergovernmental collaboration and work towards a common vision, this brief proposes the creation of an 'International Declaration on Ethical Principles for Responsible AI' (further: IDEPRAI). Ethical principles are indispensable building blocks for Responsible AI and should therefore be the foundation of the IDEPRAI¹⁰. To define the content of these ethical principles, we conducted thorough research on existing AI strategies put forward by national governments³⁴⁻⁴⁶, companies and international institutions⁴⁸⁻⁵⁸ (Annex II), in addition to academic literature review and expert consultation (Annex III). The core principles below reflect commonalities abstracted from this research, taking into account the underlying values of the SDGs. They provide the basis for both the content of the IDEPRAI and practical recommendations for the national level.

Core principles on Responsible AI

1. Global collaboration

On ethical and Responsible AI

2. Responsibility & Accountability

Concerning data security & liability

3. Awareness & Capacity

Building resilience for the impact of AI on society.

4. Data quality & Transparency

Enhancing explainability and data transparency of AI decision-making processes.

Justification

This policy brief proposes the creation of a UN Declaration that will facilitate alignment of values and implementation of policy¹⁵. It encourages countries to join the UN platform and to benefit from international collaboration, in addition to facilitating the exchange of knowledge and best practices¹. Since a declaration possesses a degree of regulatory incentivization it can provide a solid basis for legitimized national regulation on AI¹⁶. Finally, an international Declaration addresses public and private demands for a global agreement on Responsible AI.

Disclaimer

The IDEPRAI is an example of an international declaration on Responsible AI. This policy brief aims to inspire national policymakers and the international community in establishing a Declaration on Responsible AI and to provide science-based input for such a Declaration. It is not the aim of this paper to propose the exact establishment of proposed articles and content.

Towards Intergovernmental Collaboration on Responsible Artificial Intelligence

The following section discusses the outcomes of the conducted research. Structured by the four core principles, the brief addresses what the Declaration could entail and provides recommendations for their implementation. In the first section the recommendations address both the UN and national governments. The three remaining sections address national governments, since implementation of these principles should be tailored to national ambitions, norms and values. The boxes highlight examples of national implementation.

1. Global collaboration

Phase I: *establishing an International Declaration on Ethical Principles for Responsible AI (IDEPRAI).*

Although every national strategy on AI is unique, existing strategies share convergences in their ethical considerations. The UN could distil the common ground from national AI strategies and use this as additional input for the Declaration. The Declaration should also be composed in line with Human Rights values¹⁴ and SDGs¹⁷, those being international consensus-based principles.

Phase II: *implementation at the national level*

Signing the Declaration would show the intention of national governments to work towards Responsible AI. Translating this intention into action is the next step. The Declaration enables tailor-made implementations per country, according to national ambitions and values. The fast pace of AI developments demands ongoing knowledge sharing and adaptivity from policymakers. It is therefore important to monitor emerging best practices and to share these amongst countries at a UN 'Responsible AI' platform.

In order to foster international collaboration, the UN should consider:

- Establishing a 'Responsible AI' platform that fosters the exchange of common opportunities and challenges with multiple stakeholders; identifies best practices that can serve as examples for other countries and continuously aligns definitions and concerns¹. Additionally, the UN should consider acting as a global 'watchdog' for national compliance with the Declaration.
- Appointing a UN special rapporteur on Responsible AI, who follows existing and emerging national strategies, reports on best practices in countries' implementation of AI and fosters regional partnerships.
- Conducting periodical global impact assessments to monitor the impact of AI on society, using the impact on the SDGs as a point of reference²⁴.

National governments should consider:

- Appointing an AI task force that is responsible for defining a plan of action to work towards national implementation of Responsible AI policy.
- Ensuring the auditability of policy and equipping existing institutions with feedback loops to safeguard continuous improvement for implementation¹⁸.

The United Arab Emirates was the first country to appoint an AI minister³⁴. In addition, the Kenyan government established a task force on Blockchain and Artificial Intelligence that addresses 'financial inclusion, cybersecurity, land titling, election process, single digital identity and overall public service delivery'⁴⁶.



2. Responsibility & accountability

The IDEPRAI could include a section on liability and the protection of citizens' privacy and safety. Researchers and developers should be aware of their own responsibility concerning the development and use of AI applications.

National governments should consider:

- Assessing how the Declaration would fit into existing regulations and institutions. Ensuring the existence of comprehensive privacy regulations¹⁸ - according to experts, the European General Data Protection Regulation (GDPR¹⁹) serves as an example of socially responsible privacy law.
- Assessing if existing institutions have the ability to address the harms inflicted by AI technology¹⁸. Additionally, national governments should develop a legal solidarity tool to deal with risks in AI. Part of this could be the installation of an 'AI Ombudsperson'¹⁸. This could improve citizens' trust in AI.
- Determining how issues of liability should be regulated. Approach this in concert with multiple stakeholders to enhance compliance¹ and ensure that regulations do not hamper innovation.
- Incentivizing the use of tools for AI systems, including anonymised data and de-identification to protect personally identifiable information²⁰. Users must trust that their personal and sensitive data is protected and handled appropriately.
- Ensuring human agency over algorithmic decision making: clearly define the role of 'human in the loop'³¹ when using AI to increase accuracy. Assess which tasks and decision-making functionalities should not be delegated to AI systems by taking into account societal values and understanding of public opinion^{14,18}.
- Promoting toolkits for 'ethics by design', which refers to the pre-emptive consideration of ethical principles by designers of AI technology²¹.

Estonia is establishing a legal framework for AI. This framework grants a special legal status for robots. The so-called 'Robot agent' will be a legal identity apart from legal persons and private property⁴⁷.

3. Awareness & capacity

The IDEPRAI could include a section on increasing both governmental and citizen awareness. Governments need to adapt to technological development and its impact on society¹⁸. Citizen awareness needs to be addressed in order to increase resilience against possible risks of AI⁴. Additionally, experts state that capacity-building could foster citizen control on social and democratic use of AI applications.

National governments should consider:

- Supporting Research & Development on the implications and possible benefits of AI for society.
- Investing in national research talent in order to avoid 'brain drain'²².
- Developing education and training on what AI can do and what its limitations are.
- Monitoring the introduction of AI to the market and anticipating potential shifts in the labour market by offering voluntary (re-) skilling programmes to educate employees on the social, legal and ethical impact of working alongside AI¹⁸.
- Aiming for the participation and commitment of all stakeholders and active inclusion of the whole society²³. This will ensure development and design of AI that takes into account "diversity of gender, class, ethnicity, discipline and other dimensions. Hence, it fosters inclusivity and richness of ideas and perspectives"¹⁸.
- Fostering public deliberation. This should result in the co-creation of policy standards and agreements¹⁸.

In Canada, CIFAR launched an open call for 'AI & Society workshops' to involve and inform policymakers, NGOs and the broader public⁹. The 'Montreal Declaration for a Responsible Development of AI' invited the Canadian public to co-create ethical AI values through an online questionnaire⁴⁵. In Mexico, Oxford Insights and C Minds cooperated with local governments, aiming for a bottom up approach of crowdsourcing local values and needs, in order to incorporate these into a National AI Strategy⁴¹.

4. Data quality & transparency

The IDEPRAI could include a section on transparency in algorithmic decision-making, which refers to the extent to which the output of an algorithm is explicable²⁵. Experts state that since algorithms are inherently neutral yet adaptive to their input, technical transparency does not have the highest priority. Transparency of data provides insight into the input that precedes the outcome of algorithms²⁶. Since biased outcomes are due to biased input, inclusive and open data is more relevant in this context³³.

National governments should consider:

- Incentivizing users of AI applications to conduct an impact assessment before use. Moreover, periodically evaluating the impact on society using feedback loops and update the assessment possibilities of incentivisation through trust labels. In the long term, consider the potential for hard law when incentivization appears to be insufficient over time²⁴.
- Preventing biased data that can lead to discriminatory outcomes of algorithms, especially considering vulnerable groups. Acknowledging that bias is part of human nature and therefore a pertinent topic²⁶; datasets are historically bound and not necessarily inclusive.
- Gathering inclusive data that is representative of the population, by working together with multiple stakeholders¹.
- Incentivizing open access of data and technology to increase transparency, and developing methods to administer the origin and dynamics of data²⁷.

In the Netherlands, the Platform for the Information Society (ECP) launched an 'Artificial Intelligence Impact Assessment' (AIIA) with a step-by-step scheme for organizations that plan to use an AI application. These steps include questioning the measures taken to secure the reliability, safety and transparency of the application, and the justification of AI use from an ethical and legal perspective²⁴.

Conclusion

This policy brief aimed to fill two gaps: the need for intergovernmental collaboration on Responsible AI and the need to develop and implement national strategies on Responsible AI. Structured by four core principles, the brief addressed the two gaps by providing input for the UN and national governments on the possible content of a Declaration, and practical

recommendations for implementation at the national level, according to national ambitions and values. The core principles embrace four key themes on Responsible AI: global collaboration, responsibility & accountability, awareness & capacity, and data quality & transparency. With the establishment of the International Declaration on Ethical Principles for Responsible AI and consideration of the recommendations, we believe that the potential of AI can be harnessed to its fullest extent so that it will benefit the social good and facilitate the achievement of the SDGs^{1,10,14}.

Recommendations

To enhance **global collaboration** on AI, the UN should consider:

- ❖ Developing a UN International Declaration on Ethical Principles for Responsible AI.
- ❖ Installing a UN special rapporteur and a platform on Responsible AI to share knowledge and identify best practices¹.

National governments should consider:

- ❖ Executing the national implementation of the Declaration and maintaining contact with the UN rapporteur.
- ❖ Installing a national AI task force that monitors the societal impact of implemented policy and enables periodical audits and feedback loops¹⁸.

Considering **responsibility & accountability**, national governments should consider:

- ❖ Assessing, through multi-stakeholder dialogue¹, the capacity of existing institutions and regulations to address issues of liability¹⁸.
- ❖ Ensuring human agency over algorithmic decision-making and defining the role of 'human in the loop'³¹.

Considering **awareness & capacity**, national governments should consider:

- ❖ Investing in Research & Development on the opportunities and implications of AI for society.
- ❖ Enhancing participation and commitment of society through voluntary education, training and reskilling^{18,23}.

Considering **data quality & transparency**, national governments should consider:

- ❖ Incentivizing users (companies, institutions etc.) of AI applications to conduct an impact assessment before use, in addition to evaluating and adapting periodically²³.
- ❖ Working together with multiple and multilevel stakeholders to collect representative data and stimulate open access of data & technology^{1,23}.

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Annex I: Glossary

- **Artificial Intelligence (AI):** Throughout this report, the definition of AI follows the one presented by Nils J. Nilsson: “Artificial intelligence is that activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment”²⁸.
- **Responsible AI:** Integrating ethical and moral values in the use and development of AI applications, which then become democratic, socially accountable, and guarantees human autonomy and values in conformity with Human Rights and the SDGs^{1,10,14}.
- **AI Taskforce:** A governmental institution that aims to achieve implementation of ‘Responsible AI’^{1,10,14} policy by defining a plan of action, measurable goals and monitoring of these.
- **AI strategies:** Strategies, frameworks, guidelines or principles in which countries, organizations or companies state their opinion on how to achieve responsible and ethical use of AI.
- **Declaration:** Refers to multiple parties declaring their aspirations²⁴, usually in the form of soft law.
- **Algorithmic decision-making:** Decision making based on algorithms and AI systems, that can inform human decision-making³⁰.
- **Human in the loop:** Ensuring human agency. Placing importance on the involvement of humans in AI systems³¹.
- **Robot-Agent:** Granting robots a legal status. The Robot-Agent will be a legal identity on its own, apart from legal persons and private property³².
- **Brain drain:** Educated or professional people leaving a country, sector or field to apply their knowledge somewhere else, often because of better employment opportunities.
- **Data bias:** Data bias occurs when the data distribution is not representative enough of the natural phenomenon, or when biases from the real-world get perpetuated in AI systems³³. Noted should be that AI algorithms are inherently neutral, and biased outcomes are dependent on biased input.
- **Ethics by design:** The pre-emptive consideration of ethical principles during the design phase of AI technology²¹.
- **AI Ombudsperson:** A person that represents the interests of the public, monitors and promotes the implementation of the declaration, and maintains dialogue with the relevant parties.

Annex II: National strategies and private-sector approaches on AI

Following the definition of CIFARs 2018 Report: ‘Building an AI world: Report on National and Regional AI strategies’, an AI strategy entails “a set of coordinated government policies that have a clear objective of maximizing the potential benefits and minimizing the potential costs of AI for the economy and society”. The AI strategies considered in the horizon scan are strategies guided by a document, released by national governments. The countries in bold were not funded initially; they only elucidate objectives in their documents³. Other countries (e.g. Denmark, EU, France & UK) did include funding when first announced. The table presents countries and the general ethical themes that the strategies address.

Synergies - public sector

Many governments share concerns about privacy and the societal impact of AI. Convergences revolve around the themes of transparency, accountability and fairness. Common needs include the necessity of Research & Development, international and public-private collaboration, multi-sector and multi-stakeholder cooperation, empowerment through education, re-skilling and capacity-building, insight into algorithmic decision-making and the need for an AI ministry, ombudsperson or task force.

Synergies - private sector

International (non-profit) organizations establish principles that guide the development of AI according to Human Rights including privacy, data quality and anticipation to societal transformations. Common needs include the necessity to address the challenges AI could bring for societies: potential harm to privacy, discrimination, loss of skills, economic impacts, security of critical infrastructure and long term effects of social well-being.

Dealing with the research and development of AI technologies, companies are developing guidelines to assess the impact of their products internally. In general, companies state the necessity to conduct research on the implications of AI for society and to ensure that AI developments will be inclusive.

Type of document	Examples of ethical themes addressed	Specialities
National Strategies		
China ³⁴	Wants to develop laws, regulations and ethical norms that promote the development of AI. Ethics mentioned in relation to safety, openness and transparency.	
Denmark ³⁵	Mentions Corporate Social Responsibility: developing ethical recommendations for data, possibly including a data ethics code for companies’ use of data.	
Finland ³⁶	Discusses ethics in the context of openness of health data, location monitoring and the use of robots in nursing and care work. Considers defining ownership structures of developed applications.	Mentions Japan's strategy and how Finland can learn from ‘its competitors’.
France ³⁷	Has a chapter on ‘What are the ethics of AI?’, about opening the black box, considering ethics from the design stage, considering collective rights to data, ‘how do we stay in control?’ and specific governance of ethics in Artificial Intelligence.	Explicitly mentions Human Rights.
India ³⁸	Has a chapter on ‘Ethics, Privacy, Security and Artificial Intelligence’. Ethics and AI encompass: fairness - tackling the biases AI and transparency - opening the ‘Black Box’.	Gives policy recommendations for privacy issues.
Italy ³⁹	Has a chapter on ethics that discusses data quality & neutrality; accountability & liability; transparency & openness; protection of the private sphere.	
Japan ⁴⁰	Identifies ‘welfare’ as one of the priority areas, together with “productivity”, “health, medical care, and welfare”, and “mobility & “information security”. The document mentions the word ‘ethics’ once.	Second country to develop national AI strategy.

Mexico ⁴¹	Mentions ethics as a key theme in the strategy. Mentions two recommendations concerning ethics and regulations: bring data assets inside the scope of competition law, and create a Mexican AI ethics council.	Report was commissioned by the British Embassy.
Sweden ⁴²	Mentions ethics, together with safety, security, reliability and transparency.	Wants to develop rules, standards and norms.
UAE ⁴³	Document is a web page that does not mention ethics. The page does mention the need for law on the safe use of AI.	First country in the world to install a minister on AI.
UK ⁴⁴	Is establishing a Centre for Data Ethics and Innovation to advise on the ethical use of data, including for AI.	
Private-sector strategies		
Artificial Intelligence ⁴⁸ for Good Foundation	Conducts research on AI in relation to global sustainable development in order to maximize AI benefits for the social good. The foundation pleads for a common vision and global cooperation on AI.	Only strategy that mentions developing countries.
Deep Mind ⁴⁹	Looks into key ethical challenges, amongst which privacy, transparency and fairness, and governance and accountability.	
The Future of Life ⁵⁰ Institute (Asilomar Principles)	Pleads for beneficial intelligence that empowers people. Encompasses 23 principles, mainly about ethics, signed by 1273 AI/Robotics researchers, and 2541 others (by December 17, 2018).	
Google ⁵¹	Established objectives for the development and use of AI applications that are directed towards ensuring inclusivity and fairness.	
Intel ⁵²	Published a white paper on AI that provides recommendations on the ethical design of AI.	
IBM ⁵³	Developed a guide for designers and developers that focuses on ethics in five main areas: accountability, value alignment, explainability, fairness and user data rights.	
IEEE ⁵⁴	The document 'Ethically Aligned Design' aligns technology with ethics and values such as Human Rights, well-being, accountability, transparency and awareness.	
Microsoft ⁵⁵	Published a document, called 'The Future Computed', that addresses the future of AI and its implications for society. The ethics section deals with the need to ensure that AI systems are fair, reliable and safe, private and secure, inclusive, transparent, and accountable in order to mitigate the impact for society.	
Telefonica ⁵⁶	Designed an infographic that aligns ethical principles with the SDGs.	
Partnership on AI ⁵⁷	Promotes a multi-stakeholder approach to develop and share best practices. The main focus lies on responsible development of AI, including the safety and trustworthiness of AI technologies, the fairness and transparency of systems, and the intentional as well as inadvertent influences of AI on people and society.	Google, Apple Amazon, Microsoft, Facebook and IBM are part of the founding companies of a research group called 'Partnership for AI': a multi-stakeholder organization that brings together academia, civil society and companies with the aim to exchange best practices.
World Economic Forum ⁵⁸	The report 'Harnessing Artificial Intelligence' highlights benefiting humanity and the environment, in order to unlock solutions to environmental problems.	

Annex III: Consulted experts

We would like to express our sincere gratitude to the experts that agreed to contribute to the policy brief by means of an interview. Their knowledge has been of crucial importance to the accomplishment of the policy brief. Further, we would like to thank B. Alkema and D. Freijters for their review and validation of the policy brief.

Name	Institution	Position	Interview Date
Bram Alkema	Eluced	Founder	22-11-2018
Brent Barron	CIFAR	Director, Public Policy	05-12-2018
Jessica Cussins	Future of Life Institute	AI Policy lead	30-11-2018
Dr. M.V. (Virginia) Dignum	EU High Level Expert Group AI, AI Alliance the Netherlands, University of Umeå, Sweden	Professor, researcher	28-11-2018
Daniel Frijters	ECP Platform for the information society	Topteam ECP	22-11-2018
Constanza Gomez-Mont	C Minds	CEO and Founder	23-11-2018
Dr. W.F.G. (Pim) Haselager	Radboud University Nijmegen - Donders Institute for Brain, Cognition and Behaviour	Professor AI/CNS	29-11-2018
John C. Havens	IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems	Executive Director	12-12-2018
Rajesh Ingle	Connect	Managing Director	27-11-2018
Thomas Krak	UTOPIAE, University of Utrecht, University of Ghent	PhD student, Imprecise Probability Group, Intelligent Data Analysis Group	27-11-2018
Christina Martinez	C Minds	Project manager AI for Good Lab	28-11-2018
Karen S. McCabe	IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems	Senior Director Public affairs and communications, Technology Policy and International Affairs	12-12-2018
Lucien Vermeer	ICTU	Project Manager, Head policy group AI & Ethics at ICTU	02-12-2018
Dr. Angelika Voß	Fraunhofer Institut	Policy Researcher	03-12-2018

Looking Ahead

The primary aim of the synthesis report and integrated policy briefs is to provide practical and feasible recommendations to the incorporation of AI in the policy-making process. A recurring theme embedded in the three policy briefs is the essential synergy between social and governmental institutions and technological progress. In order to fully benefit from the advantages of this emerging technology, it is necessary to address related limitations and threats. Global governance mechanisms should collaborate with the technological community/lead-developers of AI in order to collectively address the development of AI for the common good.

As AI pervasively enters everyday life, its potential applications will increase and expand. Analysing the pace of AI development, current trends allow us to cautiously predict possible future developments of the applications put forth by the policy briefs.

Predicting with AI-Geospatial Mechanisms

The capacity of AI to process vast amounts of data constitutes the perfect scenario in which to develop models able to predict outcomes related to the management of space, especially in terms of land and water. Currently, AI is unable to integrate complex social variables such as power balances or cultural contexts into its processes. Consequently, this narrows its predictive capacity down to certain fields.

However, experts agree that due to the rapid development of AI, Deep Learning combined with other processes such as Data Mining will allow AI to more accurately forecast different scenarios in the near future. Furthermore, in the future, AI could comprehensively and responsibly integrate complex social variables with geospatial data in order to visualize them in multi-layered maps.

Such maps require the international community to take necessary steps towards establishing a regulatory framework mitigating the related limitations and threats of predictive AI-Geospatial systems. The development of this technology can then significantly contribute to the achievement of the Sustainable Development Goals.

Prescriptive Analytics and AI

The practices surrounding AI are in a developmental state, with vast untapped potential for its future in various sectors. Many private companies are investing in

technologies aimed at advancing their decision-making for different courses of action⁵⁵.

In order to understand the future potential of AI applications, it is necessary to understand data analytics, which fall broadly under the categories of descriptive, predictive and prescriptive analytics. *Descriptive* analytics is the simplest category which helps to summarize results. While its applications are already widely used, they are unable to guide decisions⁵⁶. *Predictive* and prescriptive analytics address both possible outcomes (what could happen) and preventative measures (what should be done). Tools such as Machine Learning fall under these categories for predicting future situations and societal needs, especially during emergencies and disasters. *Prescriptive* analytics goes one step further by recommending alternate action plans for optimal decision-making, and represents an analytical approach that can be employed in the near future for complex and time sensitive decisions⁵⁷. Such tools could benefit policy makers in terms of their decision-making, and has the potential to improve various sectors such as logistics, healthcare and education.

Towards defining the Human-Machine Interface

The disruptive effect of AI on societies, as part of the 4th industrial revolution reinvigorates the necessity to bring back discussions on political morality and ethics at the global level. Particularly, the possibility that AI could substitute human intelligence and capacities justifies many fundamental debates regarding human norms and values. Although such technology is not here yet, and AI overtaking human intelligence is still very far on the horizon, it is essential to consider how we would like to shape this reality as a global community.

Whereas it is important to acknowledge the plurality of ethical norms and standards worldwide, their variance already indicates that the global community faces arduous deliberations. This report stresses the importance of converging on underlying values to achieve a human-machine interface in line with inherent human qualities. The establishment of the proposed International Declaration on Ethical Principles for AI will be a first step towards undertaking such an effort. This report therefore urges future policy research to encompass further exploration of such a Declaration, and accompanying underlying values.

Concluding Remarks

This synthesis report in conjunction with the three individual policy briefs has investigated the relation between AI and the achievement of progress towards the six Sustainable Development Goals under the UN 2030 Agenda, in addition to addressing the core themes of “empowerment, inclusiveness and equality”. The aim of the report as a whole has been to inform the UN Division for Sustainable Development, and all interested parties, on the potential threats and possibilities that the development of AI poses for working towards the SDGs. In addition, specific recommendations aimed at harnessing the potential of AI towards effectively achieving the SDGs have been presented. With this document we hope to have informed and inspired decision-makers with ambitious ideas, and contributed towards the realization of the UN Sustainable Development Goals and the 2030 Agenda.

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Annex I: Methodology

This report and the included policy briefs were written by fifteen students from Wageningen University as an assignment for the United Nations Policy Analysis Branch, Division for Sustainable Development. The methodology encompassed both primary and secondary data collection. Primary data was collected through expert interviews whilst secondary data was collected through scientific and grey literature.

Logical framework approach

To start our project, we used the Logical Framework Approach (LFA). The LFA was primarily used to map our objectives, the data collection plan, and our assumptions. The main objective was identified as: “Providing practical and feasible recommendations and insights to the UN Division for Sustainable Development related to implications for sustainable development (in particular SDGs 4, 8, 10, 13, 16, 17) and governance of emergent, new, exponential technology clusters, focusing on AI.” Further actions to write our report and policy briefs were based on the logical framework.

Selection themes policy briefs

The selection of themes for the policy briefs took place over the first three weeks. The criteria for topic selection was based on the terms of reference (ToR), the nexus of the 6 considered SDGs, and a discussion with our contacts at the Policy Analysis Branch of the United Nations. While deciding upon our three themes we used the following criteria:

- Topics should relate to the 6 selected SDGs, and the nexus between them.
- Topics should address emergent, new, exponential technology clusters, such as AI.
- Topics should require action or cooperation on an international (UN) level.

Horizon scanning - Literature analysis

Research was conducted through the horizon scanning method. This method entails: “a systematic examination of potential threats and opportunities, with emphasis on new technology and its effects on the issue at hand.” Furthermore, scientific literature and grey literature were utilized, such literature was collected from scientific search engines or websites of relevant institutions.

Expert consultation

Identification of potential experts was based on literature, web searches and other experts' recommendations. Experts were contacted through different mediums, primarily via email, telephone or LinkedIn. Interviews were done either in person or through Skype, and were conducted in an open and semi-structured manner. As the research process progressed and became more intensive, interviews became increasingly in depth and structured. Expert lists for each respective policy brief can be found as an annex following the briefs themselves.